

Enclosure  
EPA Technical Review Comments on the City of LA's Permit Application  
"Attachment 1, Response to EPA Technical Comments and Notice of Deficiency"  
July 2020

EPA has reviewed the City of LA's ("LA" or "the City") submittal entitled "Attachment 1, Response to EPA Technical Comments and Notice of Deficiency", transmitted via cover letter dated May 6, 2020, relating to the revised Class V experimental nonhazardous (NH) permit renewal application submitted by LA (application dated December 2018). Our review focused on evaluating LA's most recent responses to EPA's questions on the permit renewal application.

EPA's numbered questions and LA's responses are in black bold italic text (historical, for reference). We've excluded sections including the associated EPA questions and recommendations and LA's responses for which we found the City's response satisfactory and for which we have no further comments/questions at this time; this includes Attachment 2, entitled "Plugging and Abandonment Plan (Updated Attachment Q and Appendix D)".

*EPA's new comments are provided in blue italicized text* within this document, *with additional questions and requests for LA in blue bold italic text*. Several *New EPA Recommendations* are designated beginning with Roman numeral X, which continues our previous numbering and *are also in blue bold italic text* and, as before, should be either specifically accepted or a clear rationale for an alternate approach should be provided for EPA's consideration.

**RESPONSE TO EPA'S TECHNICAL REVIEW COMMENTS AND NOTICE OF DEFICIENCY**

**1. Attachment A, Area of Review (AoR) Method**

*The permit application does not include details on the AoR determination. Without knowing more about how the radius was chosen, we cannot assess how the proposed operational changes might affect the AoR.*

**1.1 Please provide calculation details on the AoR determination**

The Area of Review (AoR) is the monitoring area determined by a radius  $\frac{1}{4}$  or  $\frac{1}{2}$  miles from the proposed or existing wellbore, or the area within the Cone of Influence, whichever is greater. Cone of Influence (Col) under 40 CFR 146.6 is determined to evaluate the areal extent of pressures in the injection zone that could have the potential to cause vertical migration into USDW (statutory minimum of  $\frac{1}{4}$  mile). Similarly, Radius of Fluid Displacement (RoFD) is a critical parameter for the AoR analysis as it estimates the radial distance travelled by the injected fluid during the course of injection operation (Payne and Cooper, 2018).

The Col and RoFD are calculated using equation 1 (40 CFR 46.6) and equation 3 respectively for the current injection zone "Repetto formation." Col of 942 ft (**0.18 miles**) and RoFD of 549 ft (0.1 mile) indicates injection activity will not cause any disturbance to the Plugged and Abandoned (P&A) located more than  $\frac{1}{2}$  mile away from the injection well SFI#1.

**Figure 1** shows the AoR study performed across the SFI#1 well.

$$COI = (2.25KHt / S10^x) \quad \text{Equation 1}$$

where

$$x = \frac{4\pi KH[h_w - (h_{bo} * G_s)]}{2.3Q}$$

Equation 2

$$R = \left( \frac{Qt}{\pi H \phi} \right)^{\frac{1}{2}} \quad \text{Equation 3}$$

**Table 1 Formation properties**

|                       |                                                         |       |                      |
|-----------------------|---------------------------------------------------------|-------|----------------------|
| <b>K</b>              | Hydraulic conductivity                                  | 0.36  | ft/day               |
| <b>G<sub>s</sub></b>  | Specific gravity                                        | 1.15  |                      |
| <b>μ</b>              | Viscosity                                               | 1     | cp                   |
| <b>H</b>              | Formation thickness                                     | 365   | ft                   |
| <b>h<sub>w</sub></b>  | Hydrostatic head of USDW                                | 2200  | ft                   |
| <b>h<sub>bo</sub></b> | Hydrostatic head of injection zone from lowermost USDDW | 1800  | ft                   |
| <b>Q</b>              | Injection flowrate                                      | 76807 | ft <sup>3</sup> /day |
| <b>t</b>              | Time                                                    | 7300  | days                 |
| <b>S</b>              | Storage co-efficient                                    | 0.2   |                      |
| <b>φ</b>              | Porosity                                                | 0.24  |                      |

### Formulas and parameters

#### **EPA comments:**

*The AoR has been provided, along with an evaluation (Equations 1-6), but the basis for the current injection zone pressure and other parameters are not provided, and the injection zone pressure is reported as less than the USDW pressure or head.*

*The Theis equation parameters are not sourced for their bases, and we are also not able to locate all the input values listed in the table. The pressure wave calculation assumes that gel strength is a component in the equation and does not provide the basis for 20 lb./100 ft<sup>2</sup>.*

*Some parameters are incorrect. The S value of 0.2 is for unconfined aquifers and should be reduced to 0.001 or less for unconfined aquifers; it is our understanding the injection zone is confined. The CoI would be increased to at least 2.17 miles at S = 0.001 based on our calculations and the other parameter values used by the City, the bases for which are not provided. Also, the value of net injection zone thickness, H = 3,397 ft. in Table 2 should be 365 ft. R<sub>w</sub> should be the distance to the abandoned well, 4,333 ft, not rw = 0.3 ft. Specific gravity of the formation fluid is 1.02 to 1.025 based on Table 4 on page 29 of the application, not 1.15 as stated in Table 1.*

*In addition, the calculation does not account for fault barriers, which would cause increased pressures within the AoR and a distorted AoR shape influenced by the faults, if they were sealing. If the faults are not sealing, injected fluids could migrate into the USDW.*

### **EPA questions/requests:**

**1.1-1** *In Equation 1 above, the square root is missing from the right side of the equation. This appears to be a typo as our check on the math suggests that the calculations were done with the square root. Please correct the equation or clarify.*

**1.1-2** *Please correct the parameters (H, R<sub>w</sub>, S) according to the comments above and update the calculations accordingly.*

**1.1-3** *What are the sources for the head values (h<sub>w</sub> and h<sub>bo</sub>) in Table 1; G<sub>s</sub> = 1.15 (that equates to salinity of 250,000 ppm); i.e., are they K = 0.36 ft/day; etc.?*

**1.1-4** *On what was the injection flow rate of 76,807 cf/day in Table 1 based?*

*This converts to 13,825 bpd (9.6 bpm), slightly lower than the permitted average daily injection rate of 10 bpm. This is also slightly different from the 14,440 bpd injection rate in Table 2 (which converts to the permitted average daily rate of injection of 10 bpm).*

**1.1-5** *Why was the same rate not used for both Equation 1 and Equation 3?*

*The formation thickness in Table 1 is 365 ft. However, adding the thicknesses of the 4,900 ft. sand and the 5,200 ft. sand gives 335 ft. (as shown in the data provided for question 2.2). The response to Question 6.6.1 indicates the fracture height within the target injection interval is 365 ft.*

**1.1-6** *Please clarify why the fracture height is greater than the thickness of the sand units.*

**1.1-7** *In Table 2, what is the source for P<sub>s</sub> (1,589 psi)? Is this static pressure from the hydrostatic weight of the mud column? To what depth do P<sub>s</sub> and P<sub>f</sub> correspond – is it 3,397 ft? (The P<sub>s</sub> and an H of 3,397 ft yield a pressure gradient of 0.467 psi/ft.)*

The closest active/inactive well to the SFI#1 well is at 4,333 ft (**0.78 mile**) away from the well SFI#1, which is greater than both Col and RoFD. As such a radius of 1 mile is studied from the SFI#1 well since there are three inactive wells SP LA Harbor #301, Hards Warnock#1 and Superior B-1 located 0.78 mile, 0.88 mile and 0.92 mile, respectively from the SFI#1 well. Based on the permit obtained from the Resources Agency of California Department of Conservation, Division of Oil, Gas and Geothermal Resources, the closest well SP LA Harbor #301 is a P&A well and has a cement plug at a depth of 3,397 ft. The plugging job as witnessed was approved according to the permit. Thus, for any fluid migration through this well, the pressure increase in the formation due to the injection activity must be high enough to overcome the static fluid (kill fluid) pressure within the wellbore and the gel strength of the mud as indicated by equation 4.

### **Use of mud weight and gel strength**

#### **EPA comment:**

*The gel strength is included in the pressure wave calculation; this is considered questionable in the ZEI (Zone of Endangering Influence) calculation due to a lack of robust data on how the properties of gel change over time. Critical pressure should be calculated based on fluid density differences between the injection formation and USDW.*

## Questions/requests for the City:

### 1.1-8 Please develop AoR calculations using a critical pressure based on fluid density differences rather than using the mud column and its gel strength.

The response states that the closest active/inactive well to the SFI#1 well is 4,333 ft (0.78 mile) away. There appears to be a typo as 4,333 ft is 0.82 miles.

### 1.1-9 Please correct the typo.

**Figure 2** shows the pressure increase across the radial distance from wellbore is very minimal. After 20 years of injection operation, the pressure increase in the injection formation will be only 26 psi at 4,333 ft compared to the 228-psi required for the upward fluid movement through the SP LA Harbor #301 well to a depth equivalent to the BFW depth as calculated by equation 4. Thus, the injection activity at the SFI#1 well will not compromise the USDW since the pressure increase across the radial distance is minimal and it is further guaranteed by the low pressure observed at the monitoring wells SFI#2 and SFI#4 which are significantly closer to the SFI#1 injection well.

$$P_f + \Delta P > P_s + P_g \quad \text{Equation 4}$$

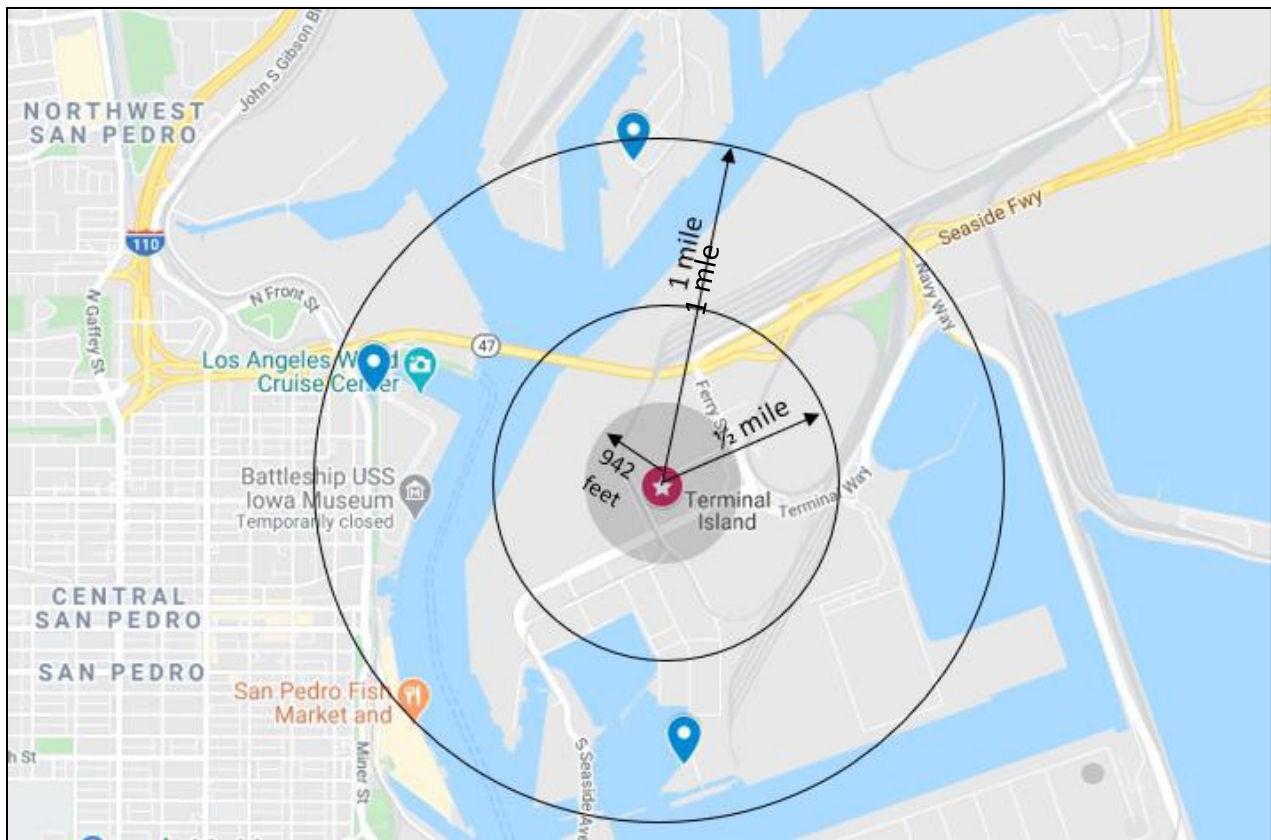
where

$$\Delta P = \frac{-70.6Q\beta\mu}{kH} \left[ \ln \left( \frac{1688\phi\mu c r^2}{kt} \right) - 2s \right] \quad \text{Equation 5}$$

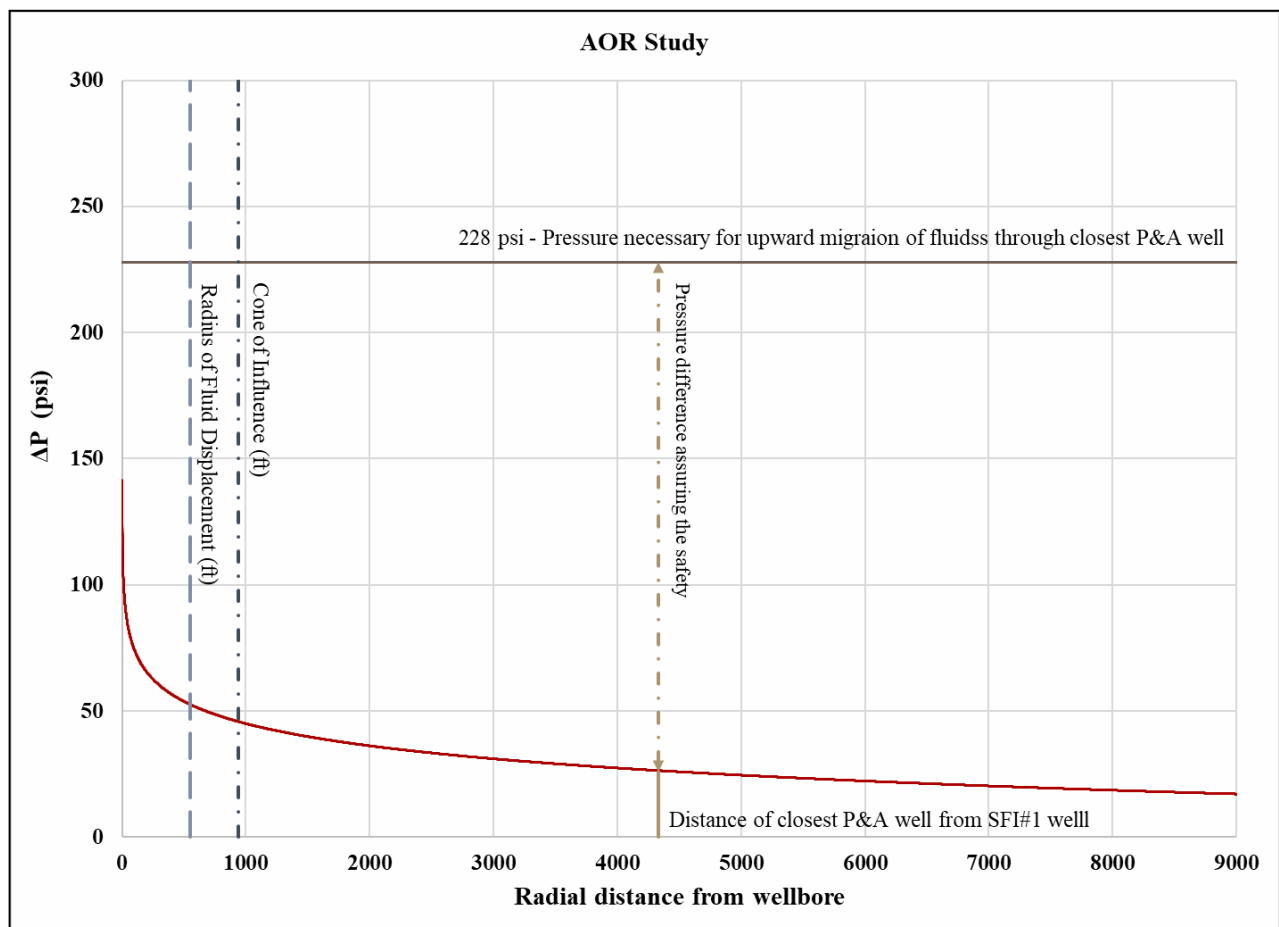
$$P_g = \frac{0.0033*G*H}{d} \quad \text{Equation 6}$$

**Table 2 Formation properties**

|                      |                            |         |                       |
|----------------------|----------------------------|---------|-----------------------|
| <b>P<sub>f</sub></b> | Formation pressure         | 2,136   | psi                   |
| <b>P<sub>s</sub></b> | Static pressure (P&A well) | 1,589   | psi                   |
| <b>Q</b>             | Injection flowrate         | 14,440  | bbl/day               |
| <b>β</b>             | Formation volume factor    | 1       | RB/STB                |
| <b>μ</b>             | Viscosity                  | 1       | cP                    |
| <b>C<sub>t</sub></b> | Compressibility            | 3.6E-06 | 1/psi                 |
| <b>r<sub>w</sub></b> | Wellbore radius            | 0.3     | Ft                    |
| <b>k</b>             | Permeability               | 100     | mD                    |
| <b>t</b>             | Time                       | 20      | years                 |
| <b>Ø</b>             | Porosity                   | 0.24    |                       |
| <b>G</b>             | Gel strength               | 20      | lb/100ft <sup>2</sup> |
| <b>H</b>             | Plugged depth (P&A well)   | 3,397   | ft                    |
| <b>d</b>             | Hole diameter              | 1.02    | ft                    |



**Figure 1: Area of Review around the SFI#1 injection well**



**Figure 2 Pressure increase along the radial distance from wellbore**

## **Results of calculations**

### **EPA comments:**

*Although the input parameters and formulas are given in this response, we have not been able to reproduce the results described for Equations 1, 3, and 5. We suspect this has to do with uncertainties in the inputs discussed above (e.g., whether the “s” in Equation 5 the same as “S” in Equation 1).*

*There is internal inconsistency in the calculated delta P values. A delta P value of 26 psi (20 years, r of 4,333 ft.) is noted above (Question 1.1) and in the response for Question 3.2. A delta P of 60 psi (20 years, r of 4,333 ft.) is referenced below in the response for Question 7.9. Also, Figure 2 shows a critical delta P of 228 psi, but the relevant calculation is not clear.*

*The calculated values are well within the proposed 1-mile AoR. Clarification of the inputs will allow us to verify that the data were incorporated properly into the calculations and to confirm the results for Equations 1, 3, and 5. Also, our calculation of the pressure increase at 1 mile is 168 psi, based on the Warner and Lehr (1981) equation and excluding gel strength pressure. The AoR could be much larger than one mile on that basis, depending on the true initial static pressure in the injection zone, unless limited by sealing faults.*

### **EPA questions/requests:**

**1.1-10 Please provide the basis for the critical delta P value of 228 psi shown in Figure 2.**

**1.1-11 Please reconcile the various delta P values: 26 psi (Question 1.1 and 3.2), 60 psi (Question 7.9) and our value of 68.5 psi.**

**1.1-12 Please provide details on the calculations that produced the results for Equations 1, 3, and 5.**

## **2. Attachment B, Maps of Area of Review**

*The one-mile AoR and its surrounding area is shown on page 11 of the application (Figure 1). A map of the project area with the locations of the two major faults is shown on page 19 (Figure 3). There are two structure maps on page 196 (Appendix F). It is not clear which units these represent, and which features are significant for injection and confinement. The 5,200- and 4,900-foot sands (shown in the structure maps) also do not appear to correspond with lithologic layers shown on the SFI#1 stratigraphic column (page 24). The Figure 3 map (page 19) includes a structure map of the upper Repetto unconformity, but is poorly labeled.*

*Core analyses are provided for SFI#1 and SFI#3 in Appendix K (page 328). The application does not include well logs or isopach maps.*

*The application does not address the current lateral distance of fluid migration in the main permit application document.*

## 2.2 Please provide isopach maps of the injection formation and confining unit.

An isopach map for the confinement is shown in **Figure 4**. The top and bottom of this isopach map are the geological interpretation for the Upper Repetto Unconformity (top) and the base of a minimum 300ft shale, below that unconformity (bottom) identified in the four SFI wells. Triangulation between the four points is applied to create the isopach map.

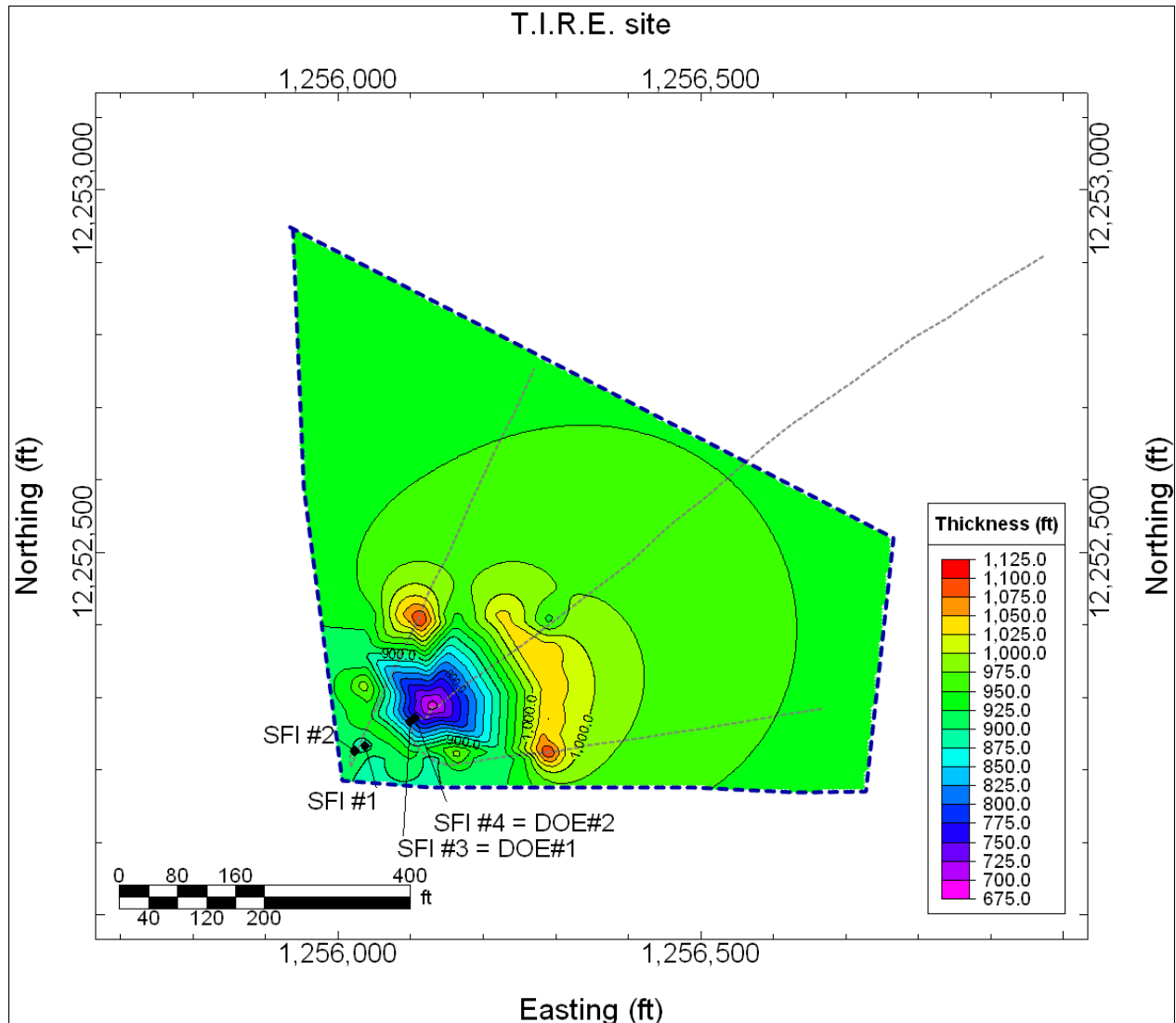


Figure 4: Isopach map confinement

### EPA question/comments:

**2.2-1** It is not clear what is meant by "...a minimum 300 ft. shale..." Is that the minimum thickness of the confining layer? The isopach map shows a minimum thickness of 675 ft.

The description indicates that the upper confinement zone is between the top of the Repetto and the base of the shale below that. According to Figure 7 in the application, the top of the Repetto occurs in the middle of a thick shale layer, and the thickness from the top of the Repetto to the base of the shale layer is about 400 ft. at well SFI#1. Extending up into the Pico, however, the total thickness of that shale sequence is closer to 900 ft., consistent with Figure 4.

The isopach map does not show the thickness of the upper confining zone to the south and west of the injection wells.

**EPA questions/requests:**

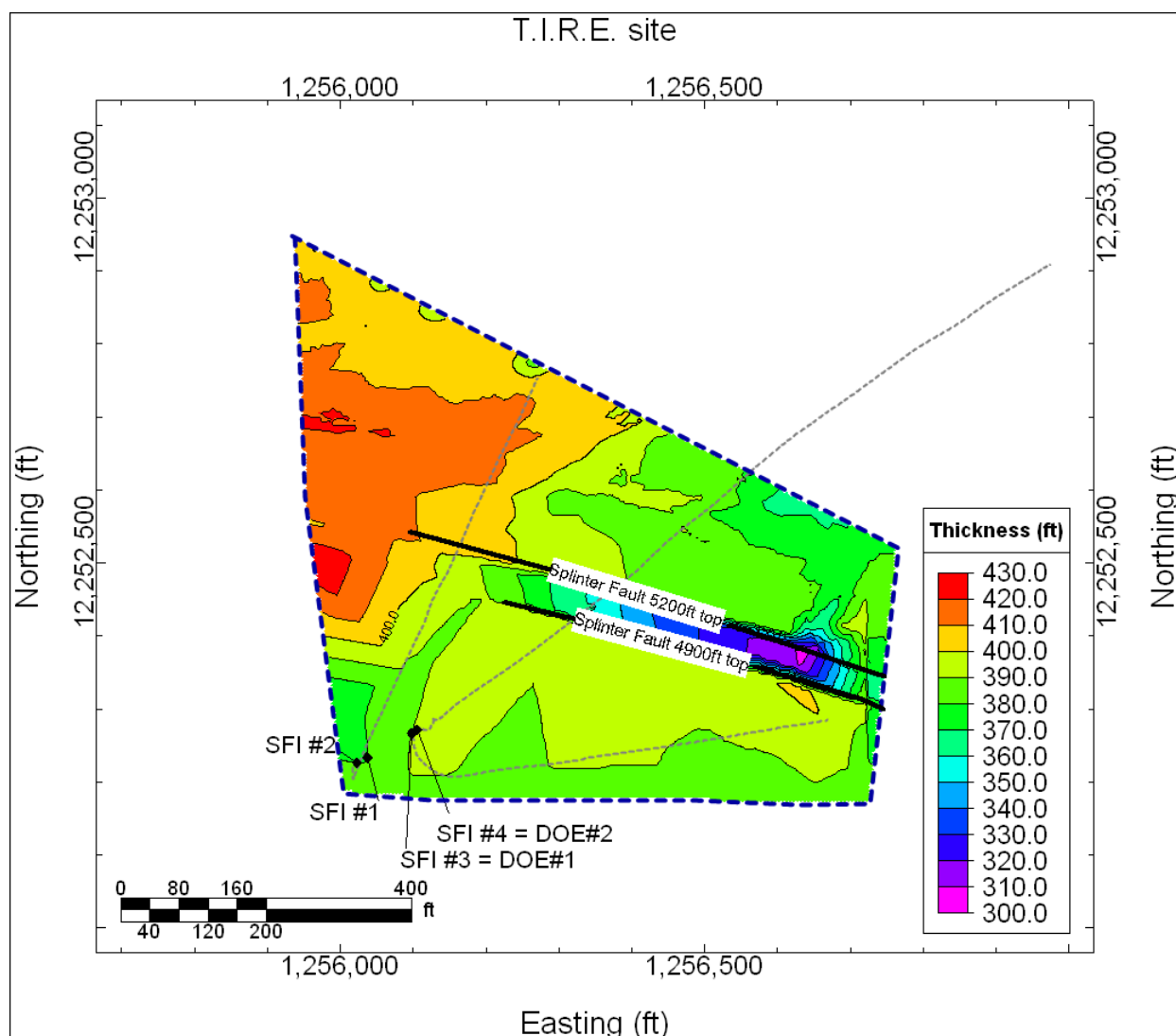
**2.2-2** Please confirm that the isopach map represents the full thickness of the shale unit that straddles the Repetto/Pico contact in the stratigraphic column (Figure 7 in application, page 24).

**2.2-3** Please provide a map that demonstrates the thickness and continuity of the confining zone to the south and west of the injection wells.

**2.2-4** What features were used to delineate the boundaries of the maps?

As for the injection formation, we want to clarify that the 4,900 ft sand structure map (page 189 of (2)) refers to the top of the sand body with thin shale interbeds identified from 4,790 to 5,065 ft in SFI#1 and similar sequences identified in the other SFI wells. 5,200 ft sand structure map (page 189 of (2)) refers to the top of the sand body identified from 5,150 to 5,210 ft in SFI#1 and similar sequences identified in the other SFI wells. Refer also to the geologic interpretation appendix J of (2))

The isopach for the injection formation is shown in **Figure 5**.



**Figure 5: Isopach map for injection formation**

***EPA comments/questions:***

*The thickness of the 4,900 ft. sand and the 5,200 ft. sand as described above total 335 ft. The AoR modeling uses a thickness of 365 ft. This is noted above in Section 1.1.*

*In the paragraph on the injection formation above, the reference to “(2)” is not clear.*

***2.2-5 Please clarify and provide that reference if not already provided.***

*The color coding on the map in Figure 5 makes it difficult to differentiate the thickness zones (i.e., the three thickness zones between 360 and 390 ft. are coded with the same shade of green).*

***2.2-6 Please provide more contrast for the different zones.***

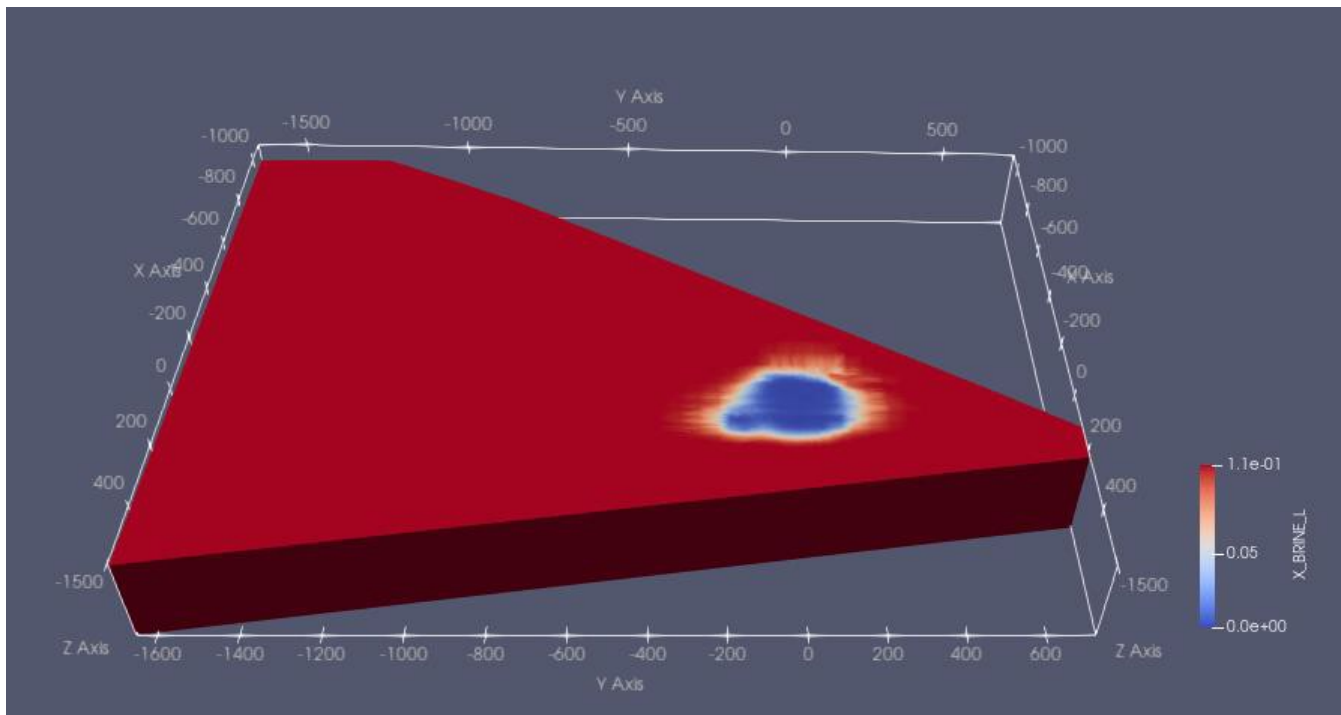
*The isopach map does not show the thickness and continuity of the injection zone to the west and south of the injection wells.*

***2.2-7 Please provide a map that shows the thickness and continuity of the injection zone to the west and south of the injection wells.***

**2.4 Please address the estimated current lateral migration of the injectate, and its rate of movement.**

The reservoir simulation performed using TOUGH2 (described in detail in section 3) estimates the injected water has pushed the formation brine to about 820 ft away from the formation after 4,266 days of injection as shown in **Figure 7**. The results obtained from the simulator for current injection and 20 years of forecasted injection is within the ¼ mile AoR as shown in **Figure 8**.

The containment of the injected fluid within the targeted injection zone is confirmed by the temperature log obtained from the fiber optic readings for SFI#1. Due to the low temperature of the injection fluid compared to the formation temperature the temperature log shows low reading for the injection zone. As shown in **Figure 9** the temperature profile shows the top of injection zone at 4,780 ft which can also be confirmed by the Gamma Ray log shown in **Figure 6**.



**Figure 7 Injected water migration after 4,266 days of injection calculated by TOUGH2**

**EPA comments:**

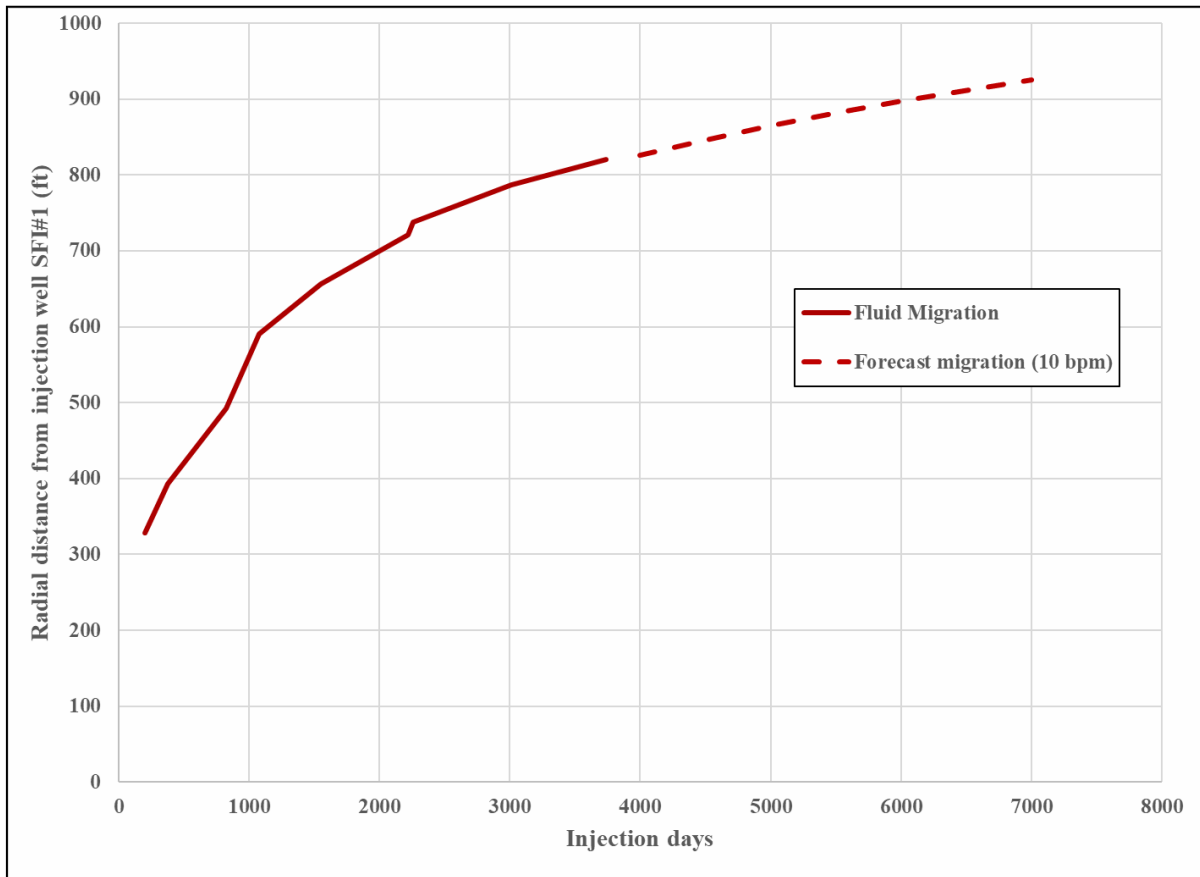
*Figure 7 is difficult to interpret as there is no north arrow for orientation, and the shape of the block does resemble the boundaries in other maps and diagrams.*

*The data in Figure 9 represent one day of testing. It appears that the temperature profile is only useful for visualizing the top of the injection formation during HPE injection; the profiles during slurry injection show a higher overall temperature and no decrease in temperature at the top of the injection zone. This suggests that the most useful temperature data would be procured during or just following HPE injection rather than slurry.*

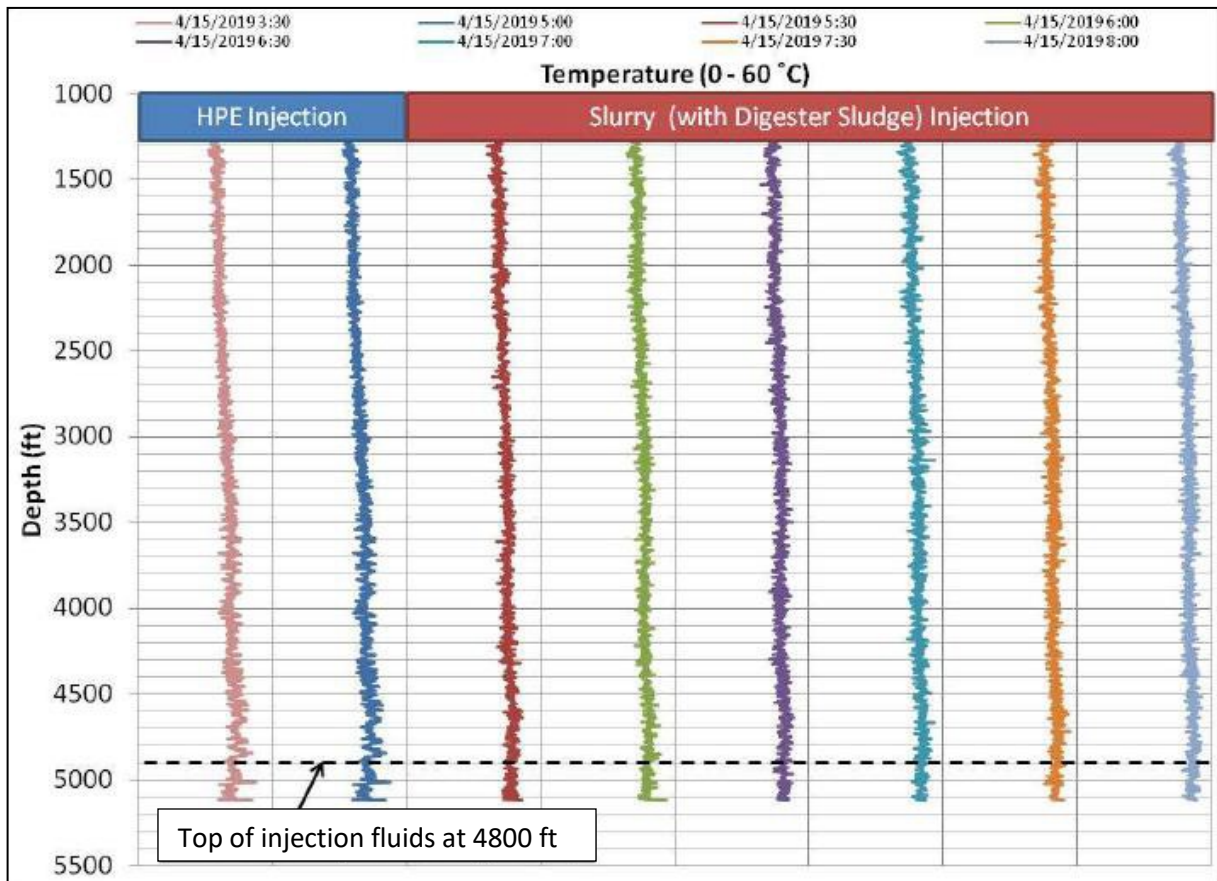
**EPA questions/requests:**

- 2.4-1** Please clarify the orientation and locations of features on the diagram in Figure 7.
- 2.4-2** Assuming the scale on the axes in Figure 7 is in feet, the plume has a diameter of approximately 400 ft., not 820 ft. Please clarify.
- 2.4-3** The City notes that the simulations indicate that injection has pushed the formation brine to about 820 ft away from the formation. What is meant by "...from the formation...?" Is this supposed to read "...from the injection well...?"
- 2.4-4** Is there any support (from monitoring wells or other field data) for the accuracy of the 820 ft. estimate in Figure 7? Please provide or describe.

**2.4-5** Since it appears that the temperature log shows a temperature drop at the top of the injection zone during HPE injection but not slurry injection, and that the temperature log during slurry injection is higher overall, how does this factor into planned monitoring using temperature logging?



**Figure 8 Radial fluid migration from well SFI#1 calculated by TOUGH2**



**Figure 9 Temperature profile showing containment of the injected fluid within target formation**

### 3 Attachment C, Corrective Action Plan and Well Data

*There are three abandoned wells located within the AoR: Superior B-1, Apex Hards-Warnock #1, and SP LA Harbor #301. Superior B-1 and Apex Hards-Warnock #1 are both plugged above the base of fresh water (BFW), which is at 2,100 to 2,300 feet below surface in the project area. SP LA Harbor #301 is plugged between the injection zone (at 3,800 to 7,500 feet deep) and the BFW. The three abandoned wells are located more than a half mile away from the SFI wells. The City has been monitoring the abandoned wells using numerical gas modeling; the application states on page 13 that “the numerical gas modeling performed within the past 7 years using SFI wells shows that the pressure did not increase more than a few psi at the monitoring well less than 600 ft away”.*

*Note that the BFW is not the same as the base of USDW (Underground Sources of Drinking Water, discussed further below). Clarification is needed on the use of that term in the application. Also, cement plugs at the injection zone and at the base of USDW are normally required as corrective action unless the monitoring data show no significant pressure effects at the improperly abandoned wells.*

#### 3.1 Please clarify the rationale for determining existing cement plug locations.

For additional security over the AoR to preserve the USDW, the critical pressure is calculated considering the fact that the offset wells were plugged above the reported USDW. The critical pressure calculation assumes a hypothetical open (bore hole) pathway between the bottom of USDW and top of injection zone. However, the critical pressure is calculated if the P&A wells falls within the Col or AoR.

Since all the 3 neighboring P&A wells are more than 3,300 ft away than the Col, the injection activity in SFI wells will not compromise the integrity of the wells and USDW in general.

***EPA comments:***

*The City appears to be setting the 1-mile AoR based on the locations of three wells in the project vicinity. (Their ZEI (Col) and waste front calculations are much smaller than 1 mile.) In their corrective action assessment, they are dismissing two of the wells without the plugs between the injection formation and the base of fresh water because they are not in the Col.*

*There are several issues with critical pressures and determination of the size of the AoR, including the Col. First, the critical pressure calculation is not shown. Also, the calculated pressure increases at the two improperly abandoned wells are well below the USDW pressure, but the basis for the current injection zone pressure and head is not provided and gel strength is improperly applied in the calculation of allowable pressure increase (see comments in Section 1.1 above). The monitoring well pressures seem to support the underpressured status, but the gauge depths are not provided, raising questions about the validity of those data.*

*Also, as noted in our comments for Section 1.1., the ZEI and waste front calculations yield a ZEI of more than one mile if gel strength is not included and the storativity value is corrected from 0.2 to 0.001 or less for a confined aquifer.*

*Ultimately, however, modeling of pressure increases in the injection zone does not replace actual measurements over time.*

***Questions/requests for the City:***

***3-1 What are the gauge depths in the monitoring wells?***

*See Section 1.1 for questions regarding Col determination and the associated parameters that are also relevant to this section.*

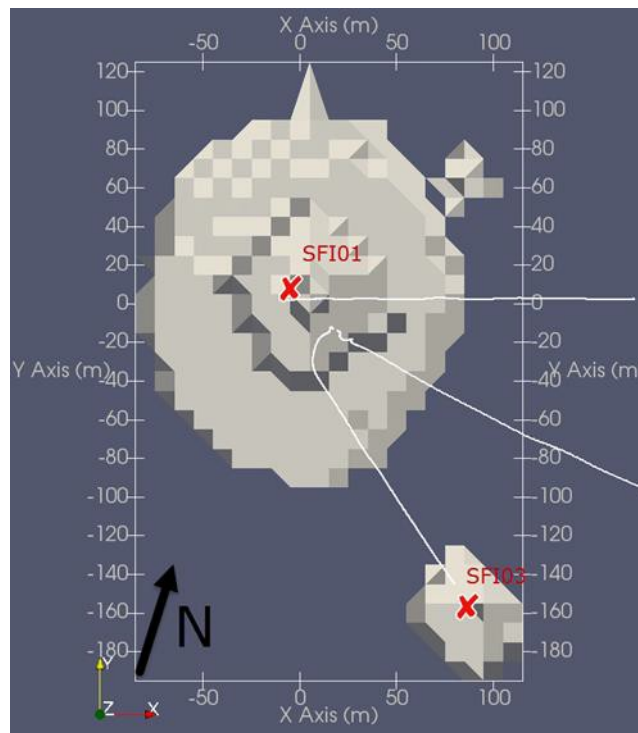
**3.2 Please provide more information about numerical gas modeling, including the model development, parameters used, calibration, uncertainties, and what exactly is meant by “a few psi” (on page 13 of the application). What are the predicted pressures at these abandoned wells, especially for the two abandoned wells without cement plugs?**

For the 10 years of operation. The injection activity does not have any effect (pressure rise) at the P&A wells. As shown in AoR after 20 years of operation the predicted pressure rises due to injection at the closest abandoned well will be 26 psi (few psi).

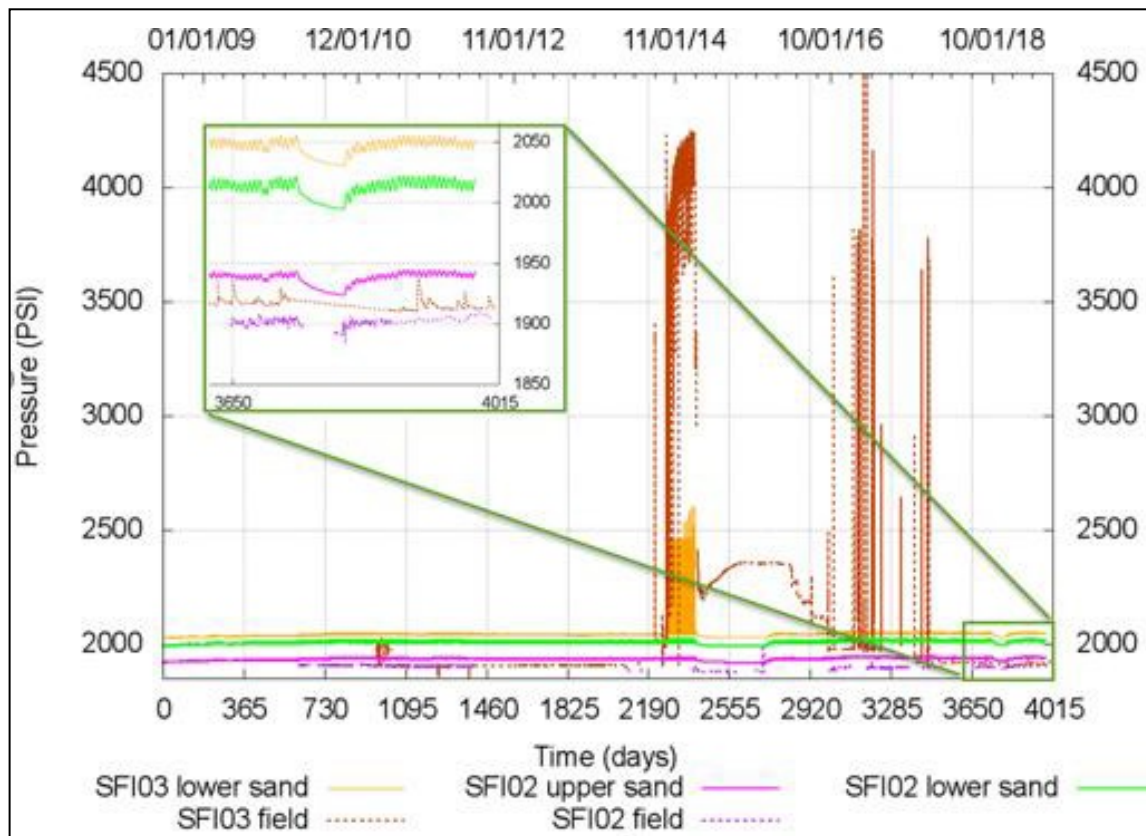
The numerical gas modeling is performed using a commercial simulation code TOUGH2 developed by the Lawrence Berkley National Laboratories. The code is used to simulate the multi-phase fluid and hear flow and transport in porous and fractured media. The model is defined with about 47,200 cells over rectangular block volume (1500m x 750m x 1200 m) oriented about N30E, with bedding planes dipping an average 20 degrees downward to the SW. A plane of symmetry through the injection well is assumed along with pressure boundary conditions to the NW and SW with a no flow boundary condition to the NE (due to the sealing character of the Thums-Huntington Beach Fault).

The simulation run for the 3,983 shows faster migration of injected fluids in the NW, SW and SE direction compared to the NE direction where the Thums-Huntington Beach Fault is present. It further indicates the accumulation of the gas plume (through degradation of biosolids) near the SFI#1 well in the 4,900 ft sand. The plume size containing minimum gas saturation of 3% extends about 100 m from the injection well (SFI#1) as shown in **Figure 10**. It further shows the accumulation of small gas close to the SFI#3 well due to an injection activity conducted from September 2014 through January 2015 in addition to the Step-Rate Test conducted using digested sludge in 2017.

Also, the simulation results indicate the pressure behavior at both monitoring wells SFI#2 and SFI#4 is consistent with the field data and does not show any significant rise due to the ongoing injection activities as shown in **Figure 11**. The pressure behavior at the monitoring wells further confirms the integrity of the neighboring P&A wells as no pressure increase has been observed at monitoring wells due to the injection activity.



**Figure 10 Top view gas plume with minimum 3% gas saturation after 3983 days of injection**



**Figure 11 Pressure at monitoring well SFI#2 and well SFI#3**

**EPA comments:**

The response does not include important information about the model such as its parameters, calibration, uncertainties, or a cross section showing model layering. It is also unclear from the response whether Figure 11 represents both field and simulation data or simulation results alone. The City states that Figure 11 indicates pressure has not increased at SFI#2 and SFI#4 due to operations, but Figure 11 shows only SFI#2 and SFI#3.

**EPA questions/requests:**

- 3.2-1** Please provide information about the model parameters, calibration, uncertainties, domain, and layering. Also, how were the faults represented in the model?
- 3.2-2** Why was a 3% gas saturation chosen for the model?
- 3.2-3** For Figure 10, please provide a legend that explains the different shades of white and gray that represent the plume.
- 3.2-4** Please clarify whether Figure 11 displays field data or model results. Furthermore, what is represented by the orange vertical spikes in the graph? Do these correspond with injection operations? Please clarify the legend.

**3.2-5 How was fracture development incorporated into the modeling since the fracture network is not static?**

**3.3 Are fall-off test results available that provide static pressure data over time? If so, how does this data compare with predicted pressures at the abandoned wells and the original AoRevaluation?**

Static pressure data are determined from the bottomhole pressure measurements of the monitoring wells SFI#2 (1,900 psi) and SFI#4 (1,839 psi), which are located within 600 ft from the injector. No pressure rise was observed at these wells due to the injection operation in SFI#1. This observation is consistent with the pressure prediction from TOUGH2 simulator as shown in **Figure 11**.

**EPA comments:**

*The response implies that Figure 11 displays model results. If this is the case, no observed data at SFI#2 or SFI#4 has been provided. See comments on response 3.2 for questions on Figure 11.*

**EPA question:**

**3.3-1 Please provide measured pressure data for SFI#2 and SFI#4.**

**3.4 Please clarify your use of the BFW term, particularly how the BFW compares to the deepest water containing less than 10,000 mg/L TDS (total dissolved solids).**

The BFW refers to Base of Fresh Water and has < 3,000 mg/L of TDS whereas the permit mentions 10,000 mg/L TDS. The USDW as defined in Title 40, Code of Federal Regulations (40 CFR), Section 144.3 is an aquifer or part of an aquifer which aquifer supplies any public water system, or contains a sufficient quantity of ground water to supply a public water system and currently supplies drinking water for human consumption or contains fewer than 10,000 milligrams/liter of Total Dissolved Solids (TDS)

The term BFW was used because the salinity decreases with depth as we move from seawater zone to freshwater zone, at depths deeper than 2,100 an increase in the water salinity will be observed again as per the paper referred in permit to show the hydrological unit for the Wilmington Oil Field by Henderson, (1987). The hydrological unit shows the BFW is within Pico formation which is confirmed by the resistivity log within Pico formation as shown in **Figure 12**. The resistivity log indicates that the deepest water source is at 2,100 ft.

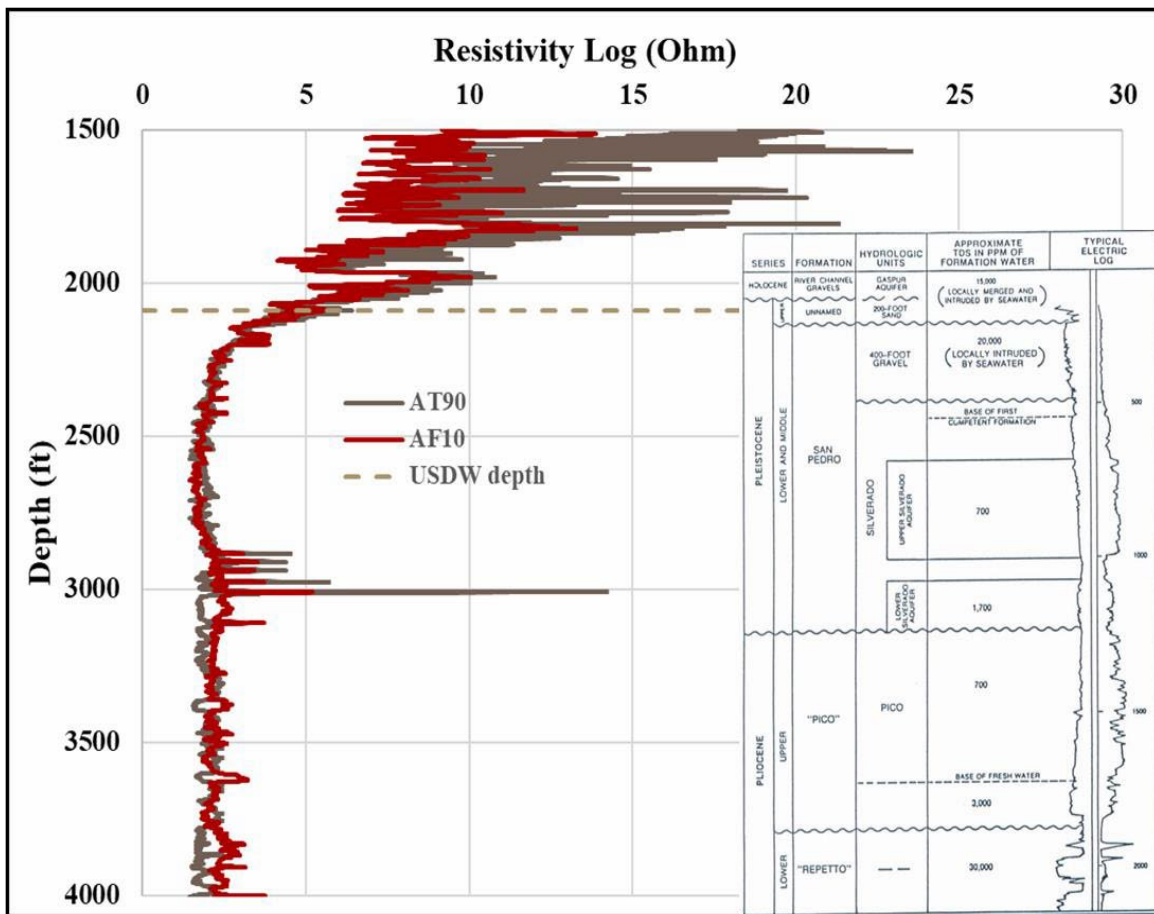


Figure 12 Resistivity log of SFI#11 well and Hydrological unit of Wilmington Oil Field (Henderson, 1987)

**EPA comments:**

The response states that the resistivity log and the BFW in Figure 12 are within the Pico Formation, but the referenced stratigraphic column would place the BFW in the Repetto Formation. The vertical scales do not seem to match.

**EPA questions/requests:**

- 3.4-1 Please clearly identify the base of USDWs or confirm that the BFW is equivalent to the USDW.
- 3.4-2 Please explain the discrepancy between the two vertical scales shown in Figure 12 and identify in which formation the BFW is situated.
- 3.4-3 Figure 12 shows a resistivity log of SFI#11. Assuming this is a typo, please correct the figure to SFI#1.

## 5 Attachment F, Maps and Cross Sections of Geologic Structure of Area

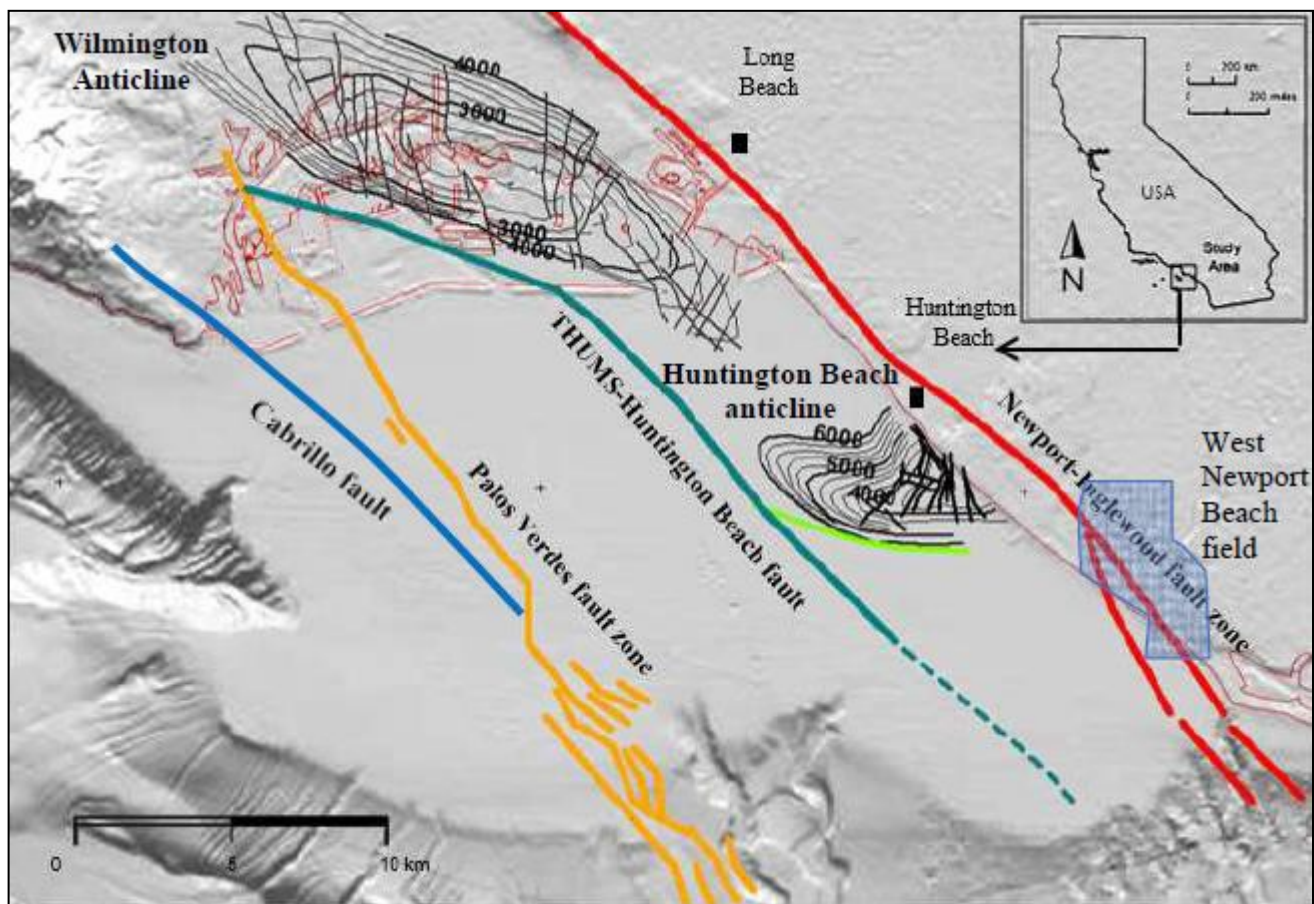
### Faults and Fractures

As shown in the Figure 3 project map, the project is located between two major faults: the Thums-Huntington Beach and the Palos Verdes faults. The Palos Verdes is a fault zone about 300-1000 feet

wide. It is nearly vertical with a slip rate of about 0.3 to 0.4 mm per year (page 19). By contrast, the application states on page 19 that “little is known about the Thums-Huntington Beach fault.” As shown in the Figure 5 and 6 cross sections, all four SFI wells cut the Thums-Huntington Beach Fault (page 20). In Don Clarke’s geologic analysis in Appendix J, the application mentions that one estimate of the fault’s dip is 55 degrees NE (page 294) and it does not appear to have displacement in this area (page 301 and Appendix J Figure 7). The two faults are converging (page 312, Appendix J). The application does not address whether these faults are sealing.

**5.1 Is there evidence to suggest that the Thums-Huntington Beach and Palos Verdes faults are sealing faults? If so, please provide and explain this data.**

The THUMS-Huntington Beach fault (T-HBF) is located between the two major fault systems (The Newport-Inglewood fault zone (NIFZ) and Palos Verdes fault zone (PVFZ) subparallel strike-slip fault) and may merge with and transfer slip to or from both, but tectonic relationships among these fault zones have not been clearly established. The PVFZ is a dextral, strike-slip system with a reverse component that strikes southeast and extends from Palos Verdes Hills onto San Pedro Shelf, forming the southwest boundary of the Wilmington Graben (WG; Figure 2; Francis and Legg, 2007; Fisher et al., 2009; Sorlien et al. 2013). The T-HBF forms the northeast boundary of Wilmington Graben and the southwest boundary of Wilmington and Huntington Beach anticlines as shown in **Figure 13**. Therefore, a possibility exists that the fault could have affected oil migration and trapping in the Wilmington and Huntington Beach fields (Ishutov, 2003). Due to the trapping mechanisms of the fault the upward migration of the fluids is prevented making it a sealing fault.



**Figure 13: Base map showing location of Wilmington and Huntington Beach anticlines and NIFZ (red) and T-HBF (green) are from Wright (1991). PVFZ location (orange) is from Francis and Legg (2007)**



**EPA comment:**

*The response reasonably argues that, because the THUMS-Huntington Beach fault separates the Wilmington graben from the Wilmington and Huntington Beach oil fields, it is likely to be a sealing fault. The response does not, however, address whether the Palos Verde fault zone is sealing. Ultimately, the relationships between these faults is still not well characterized.*

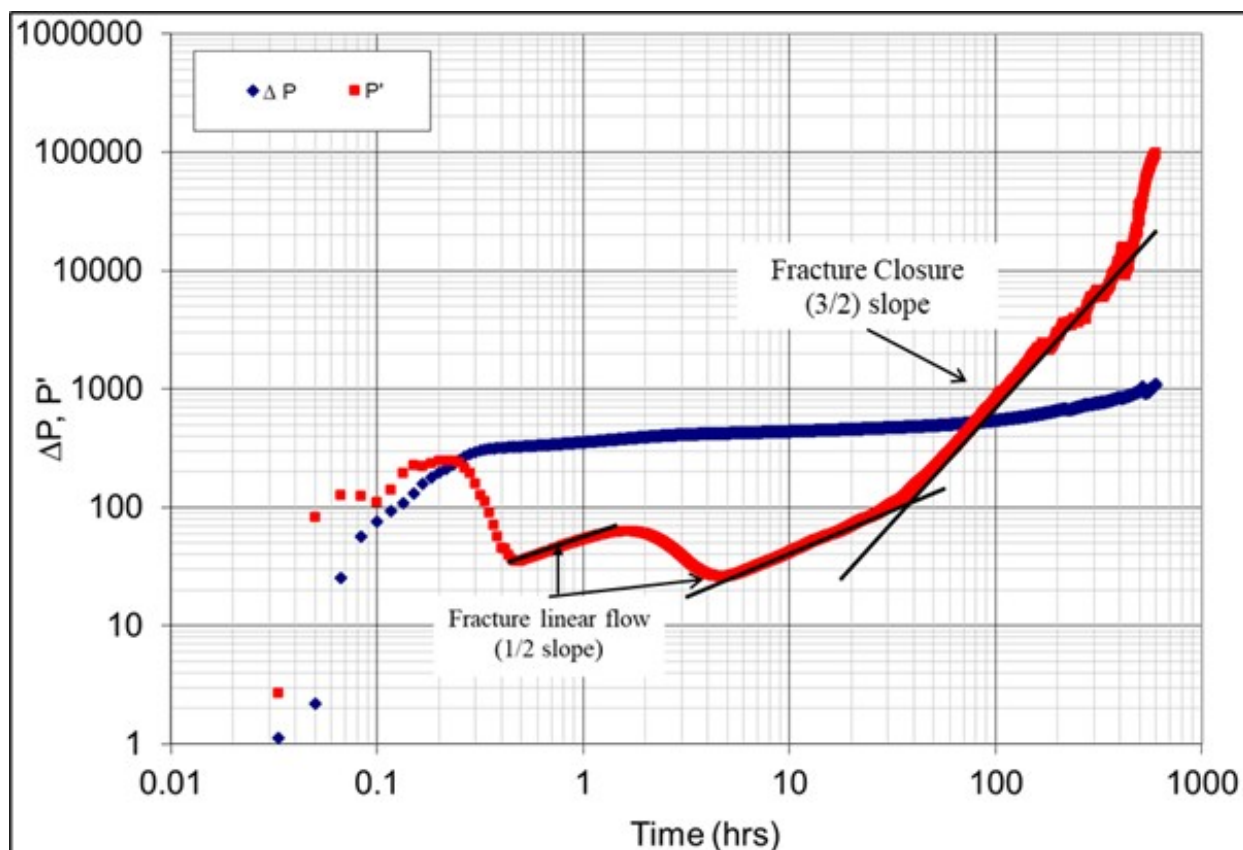
**EPA request:**

**5.1-1 Please discuss whether the Palos Verde fault zone is likely to be sealing.**

**5.2 Is there evidence of fluid migration along these faults, or any operational data that would suggest future migration is possible? Please provide and explain this data.**

There is no evidence of fluid migration along the faults. The geological data provided in the response to question 5.1 suggested that the fault is sealing. Also, the reservoir simulation model presented in the response to questions 2.4 and 3.2 show that the injectate has not reach to the fault to date.

The pressure fall-off analysis performed during the elongated shut-in period in 2018 is shown in **Figure 14**. The pressure analysis did not capture boundary signature, which usually indicated by late unit slope in pressure derivative diagnostic plot. From this we can conclude that the pressure signal due to injection operations has not reached the boundary, which eliminates the risk of injected fluid migration through a leaky fault if any.



**Figure 14 Pressure fall off analysis (log-log plot) using the elongated shut-in data**

### **EPA comments:**

*The response focuses solely on the THUMS-Huntington Beach fault and does not address the Palos Verde fault zone. It is difficult to confirm that the reservoir simulation model indicates that the injectate has not reached the fault because no map of the model domain showing the faults has been provided.*

*The City states that because the pressure fall-off analysis shown in Figure 14 does not display a boundary signature, the injected fluid has not reached a hydraulic boundary (i.e., the THUMS-Huntington Beach fault).*

*Regarding the FOT data, the Region 6 guidance on FOTs indicates that a late increase in slope could indicate reaching multiple faults that bound a reservoir. Although not conclusive, that interpretation could be consistent with the furthest portion of the plot shape in Figure 14.*

### **EPA questions/requests:**

**5.2-1** *Please address the potential for fluid migration along the Palos Verde fault zone.*

**5.2-2** *Please add the fault locations on the map of the modeling results.*

**5.2-3** *In what well was the Figure 14 fall-off analysis performed? If a full report of that FOT is available, please provide a copy.*

**5.2.4** *Please provide any evaluation by the City regarding what may happen if injected fluid reaches the boundary, since the suggestion that the THUMS-Huntington Beach fault is sealing (see response 5.1) is still speculative. Similarly, please share any evaluation of the potential for formation fluids to migrate through the fault if it is not sealing and intersects the Col.*

### **Cross Sections**

*Section F of the application provides computer simulated cross sections of the project site (Figure 3 and Figure 4; application Figures 10 and 11). They show the location of the Thums-Huntington Fault, the locations of the SFI well paths, and the lithology between approximately 3,750 and 5,500 feet. A location map for the cross sections can be found in Appendix F, page 197 of the application.*

*The cross sections do not show the full extent of the injection zone or either of the confining layers. The cross sections need to clearly label the injection zone and upper and lower confining units. The application also includes a cartoon diagram of the chevron folds in the project area's fault block, taken from Don Clarke's geologic report in Appendix J (Figure 5; application Figure 6 on page 20):*

**5.3 The cross sections in Figures 5 and 6 changed since the previous permit application. Please provide data used to generate these computer-simulated cross sections.**

Figures 5 and 6 show the lithology cross-sections along different well paths. The lithology for the T.I.R.E. site was compiled from the nearby Wilmington oil field and refined by the drilling of the four (4) SFI wells. The lithology cross sections in the previous permit application dated in the Year of 2013 were based on the refinement using the log data including mud log, GR, and Resistivity log of Well SFI#1, SFI#2, and SFI#3, however, the deepening of Well SFI#1 (from 5550 ft to 7039 ft) and drilling of the latest Well SFI#4 (TD of 7650 ft) occurred in the Year of 2014. SFI#1 and SFI#4 are bottomed in the Miocene Puente Formation. Therefore, the lithology model and cross-section maps were modified

based on the data on the latest drilled wells and reflected those changes in the new application document.

The data used to generate these cross-sections are the lithology interpretation (based on GR, resistivity, and mud log) at the SFI wells, average dipping of the three domains based on geological interpretation – see Fig. 3 page 291 in the application file - and historical nearby Wilmington oil field log data.

The original log data (mug log, E-log) and the interpreted lithology columns for the four (4) SFI wells can be found in the attached zip file “Lithology MudLog ELog”. There are two folders under this zip file. The “Lithology” folder contains a spreadsheet containing the interpreted lithology columns for the four SFI wells, while the “Logs” folder contains the E-log and mud log for the four SFI wells. The list of files is shown in **Table 3**.

**Table 3: Data files for lithology model preparation**

| Folder                     | Filename                     |
|----------------------------|------------------------------|
| Lithology                  | LithologyInterpretation.xlsx |
| Logs/SFI01/Deepening       | SFI #1_ELog_Deepening.las    |
|                            | SFI #1_MudLog_Deepening.las  |
|                            | SFI #1_MudLog_Deepening.pdf  |
| Logs/SFI01/OriginalShallow | SFI#1_Elog_Shallow_1.las     |
|                            | SFI#1_Elog_Shallow_2.las     |
|                            | SFI#1_MudLog_Shallow.las     |
|                            | SFI#1_MudLog_Shallow.pdf     |
| Logs/SFI02                 | SFI#2_ELog.las               |
|                            | SFI#2_MugLog.pdf             |
|                            | SFI#2_MugLog.las             |
| Logs/SFI03                 | SFI#3_ELog_1.las             |
|                            | SFI#3_ELog_2.las             |
|                            | SFI#3_MugLog.pdf             |

**EPA comments:**

*Because the lithology was compiled from the Wilmington oil field, which is located on the other side of the THUMS-Huntington Beach fault in the Wilmington anticline, the City should discuss how the logs from that field can be correlated with the project site.*

**EPA questions/requests:**

**5.3-1** *Please discuss how the lithology of the Wilmington oil field can be correlated with the logs from the project site, given that they are separated by the THUMS-Huntington Beach fault.*

5.4 Please provide cross sections that cover the full depth of the project area and clearly label the injection zone and confining units. Cross sections should include a horizontal scale.

Figures 15 through Figure 18

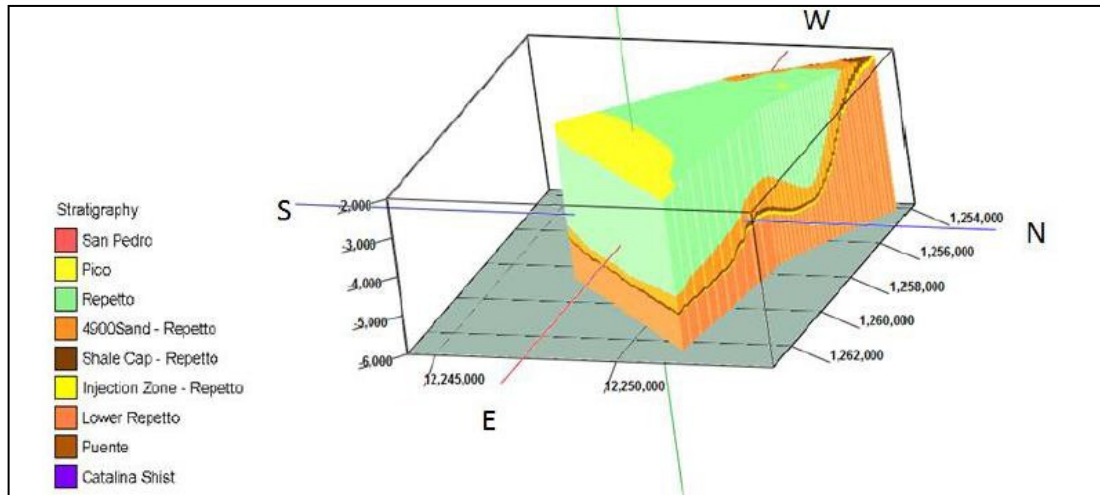


Figure 15 Cross section below TIRE

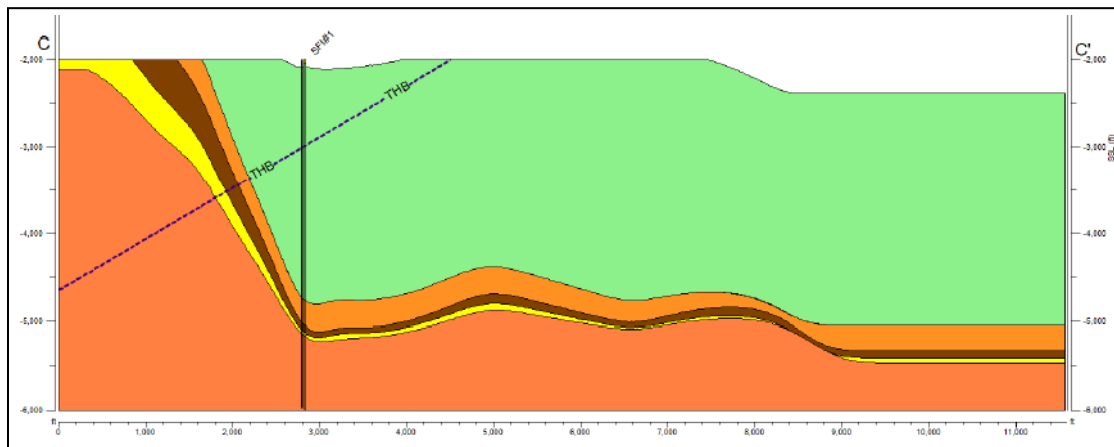


Figure 16 Cross section C C'

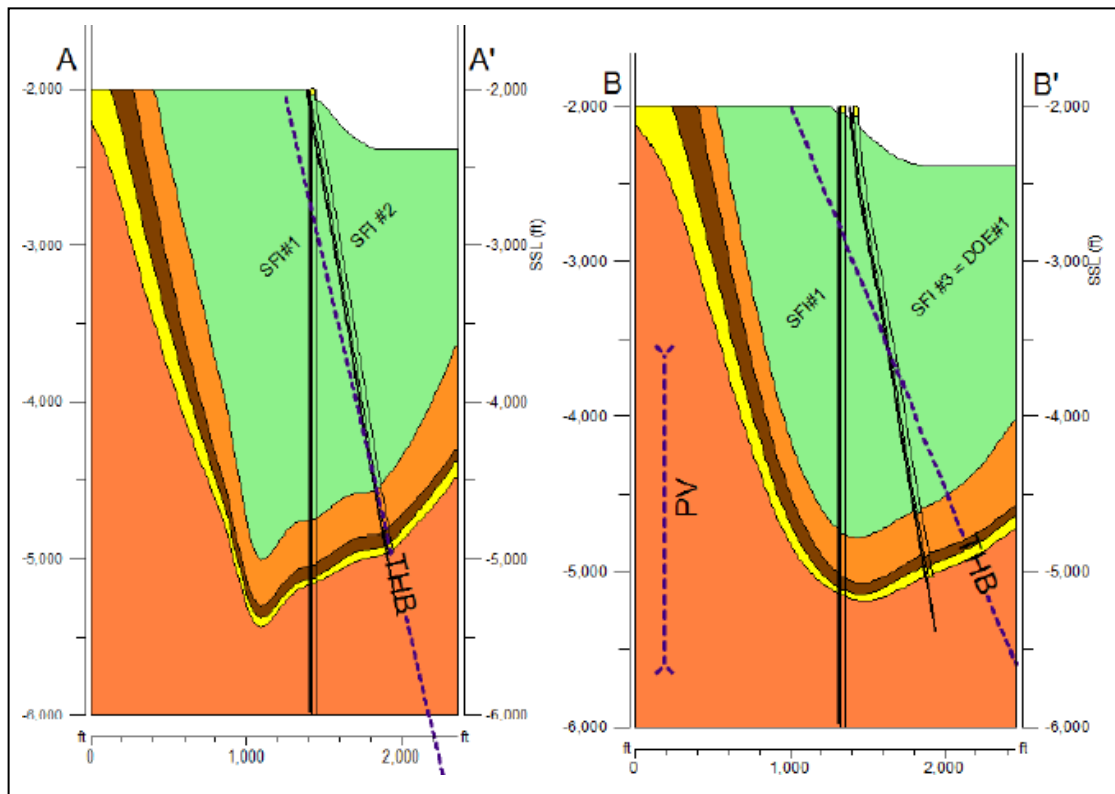


Figure 17 Cross section A A' and B B'

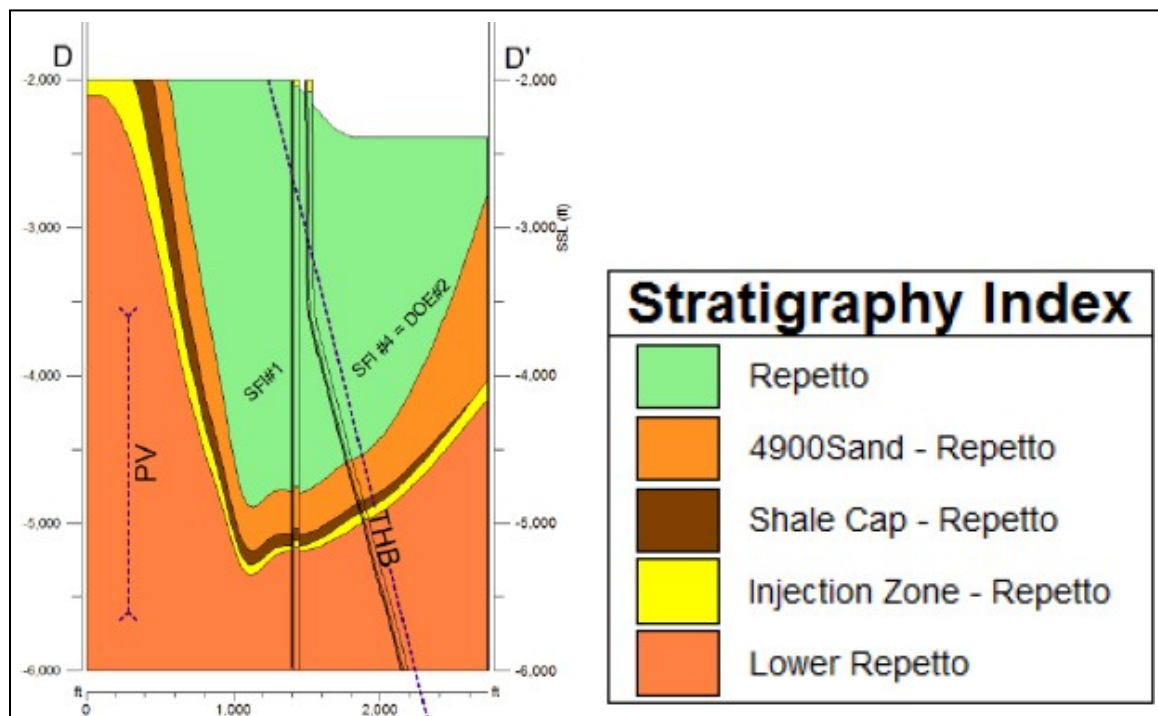


Figure 18 Cross section D D'

**EPA comments:**

*The figures have been provided, but they are missing a regional index map with the locations of the cross sections marked. There is also some inconsistency between the cross sections shown here and the description of the units in the application.*

**EPA Questions/requests:**

**5.4-1** *Please provide a regional map key with the locations of the cross sections.*

**5.4-2** *Please provide a key to the horizontal scale in Figure 15.*

**5.4-3** *The cross sections show an injection zone in the Repetto Formation, but not in the Pico Formation. Are there any other planned injection zones not shown in the above cross sections?*

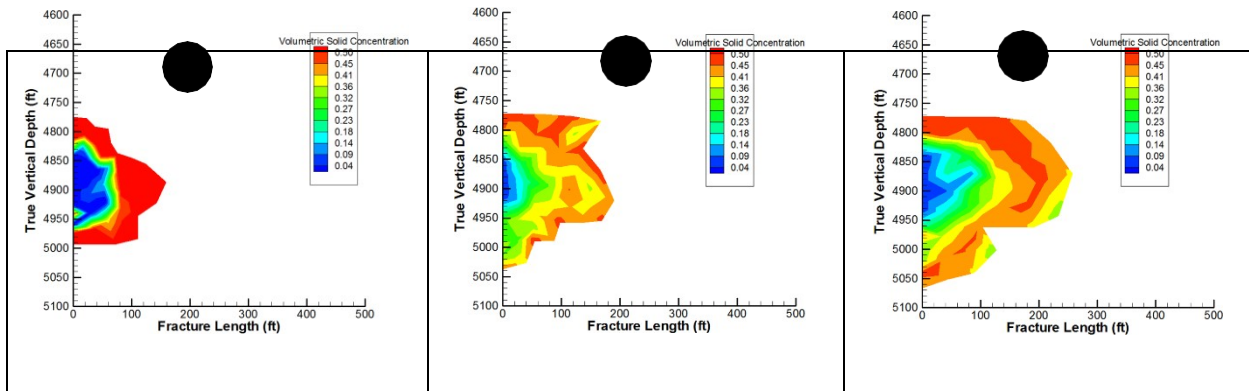
**5.4-4** *The application describes the primary upper confining layer as an 800 to 1,000-foot-thick shale found at depths between 2,100 and 3,200 feet in the upper Repetto Formation and lower Pico Formation. This is not shown in the cross sections. Please explain this inconsistency.*

**6. Attachment G, Lithology and Stratigraphy for SFI #4**

The fracture simulation conducted for SFI#1 (**Figure 29**). **Figure 29 A** shows at injection flow rate of 10 bpm (lower than the fracture flow rate), a solid packed hydraulic fracture was created. The maximum fracture half-length achieved is 160 ft. Furthermore, the study showed that a longer fracture will be created with much lower solid concentration inside the fracture at an injection flow rate higher than the fracture flow rate (20 bpm). If the injection volume is kept at the current average volume (~ 9,000 bbl/day) the created fracture half-length in the higher rate scenario is 200 ft as shown by **Figure 29 B**. Extending the injection volume to 18,000 bbl will extend the fracture half-length to 260 ft as shown by **Figure 29 C**.

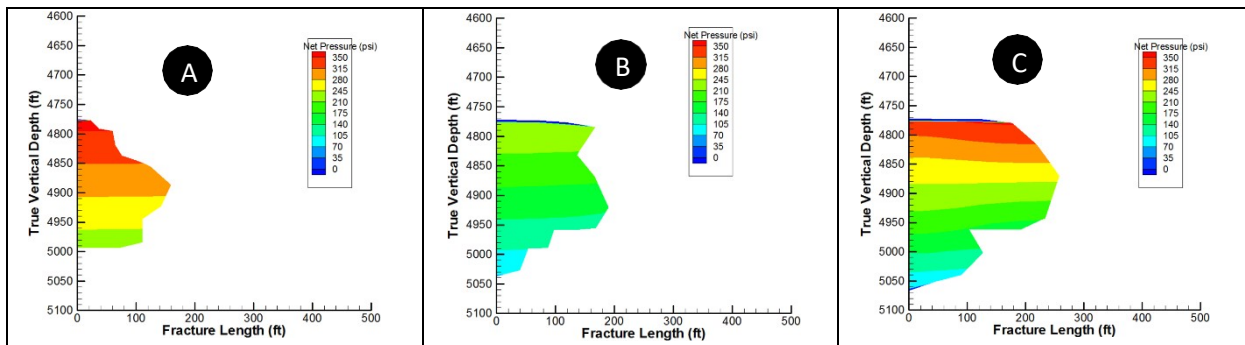
The net pressure (pressure above formation stress) inside the fracture was also calculated by the simulator to predict the injection pressure at the higher injection flow rate. The estimated maximum net pressure at the current injection flow rate (10bpm) is 350 psi as shown by **Figure 30 A**. Based on the estimated net pressure value and the fracture pressure obtained from the SRT and geomechanical analysis (4050 psi) the calculated injection pressure at 10 bpm is 4,400 psi, while the current injection pressure ranges between 4,350 and 4,400 psi. Increasing the injection flow rate will result in a slightly lower net pressure as lower solid concentration inside the fracture was estimated as shown by **Figure**

**30 B, and 30 C.** The estimated net pressure at 20 bpm ranges between 240 and 330 psi, which will result in injection pressure of between 4,290 and 4,380 psi.



**Figure 29 Fracture simulation results (SFI#1), Solid Concentration**

**A) Q = 10 bpm\_V = 9,000 bbl    B) Q = 20 bpm\_V = 9,000 bbl    C) Q = 20 bpm\_V = 18,000 bbl**



**Figure 30 Fracture simulation results (SFI#1), Net Pressure**

**A) Q = 10 bpm\_V = 9,000 bbl    B) Q = 20 bpm\_V = 9,000 bbl    C) Q = 20 bpm\_V = 18,000 bbl**

**6.5 What evidence is there that far field pressure has not significantly increased, despite the skin effect caused by the fibrous biosolid injectate? Please provide and explain.**

The static pressure data are determined from the bottomhole pressure measurements of the monitoring wells. The values for both monitoring wells, SFI#2 and SFI#4 are 1,900 psi, and 1,839 psi, respectively. Both wells are located within 600 ft from SFI#1.

**EPA comments:**

*The response does not provide historical pressure measurements for SFI#2 and SFI#4. This information has been requested in the comments for Response 3.3, above.*

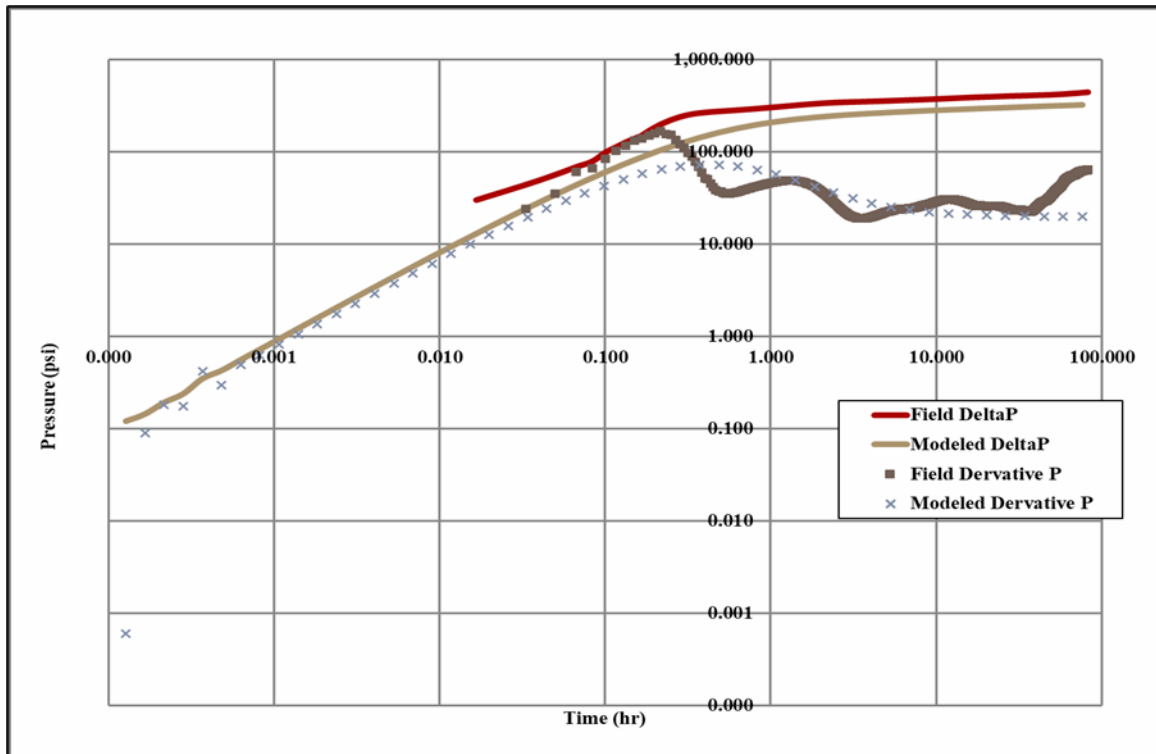
**6.6 What are the porosity and permeability of the shales in the Pico Formation?**

The average permeability of Pico formation is less than one-tenth that of the San Pedro (Upson and Thomasson, 1951). However, the horizontal permeability ( $K_h$ ) is generally greater than vertical ( $K_v$ ), so the permeability of Pico formation is quite low and is considered impermeable.

The porosity of Pico formation was determined from the porosity log to be 28%. This shale formation is not permeable as confirmed by no upward fluid migration. Moreover, it was also confirmed by the temperature log data and special well test analysis (please refer to the "Lateral extension of the injection fracture (Figure 9)").

Additionally, a special pressure fall-off analysis is conducted to confirm the containment of the fracture within the targeted injection zone. Type curve matching with different reservoir models has been applied to analyze the pressure fall-off data using an advanced commercial well testing software (@IPT) described in greater detail in Section 10.11.

**Figure 31** presents the results of special fall off analyses. The figure shows an excellent match between the pressure change and its derivative with the corresponding simulated curves. The match represents a reservoir system where injection fluid flows through a single vertical hydraulic fracture propagated from the well inside an inner ring that has properties different than the far field formation properties. The fracture and reservoir properties captured are summarized in **Table 4** and it confirms the containment of fracture within the targeted injection interval with fracture height of 365 ft.



**Figure 31 Type-curve match and results for falloff test in SFI #1 well in March 2020**

**Table 4 Results of fall-off test analysis**

| Parameter                                 | Value |
|-------------------------------------------|-------|
| Inner Zone Permeability (mD)              | 35.58 |
| Fracture Half Length (ft)                 | 37.50 |
| Fracture Storage Constant (bbls/psi)      | 0.6   |
| Injection Layer stress (psi)              | 4234  |
| Containment Layer Stress(psi)             | 4434  |
| Length Shrinkage Speed Parameter (Delpat) | 2.8   |
| Mobility Front, Elliptical                | 1.34  |
| Mobility / Diffusivity Ratio              | 0.26  |
| Outer Zone Permeability (mD)              | 134   |
| Fracture height (ft)                      | 365   |

**EPA comments:**

*The reference Upson and Thomasson (1951) does not appear to address either the Pico Formation or the San Pedro Formation, so it is not clear what the permeability of the Pico Formation is. The response also does not include the porosity log that found a porosity of 28% for the Pico Formation.*

**EPA questions:**

**6.6-1**      *What is the permeability of the Pico Formation and how it was determined?*

**6.6-2**      *When and in which well was the porosity log performed? Please provide the log.*

*The application states on page 23 that the BFW in the project area is 2,100 to 2,300 ft, in the Pico Formation. This is entirely within what the application identifies as the confining shale.*

**6.7 Please clarify or discuss how injection activities will be protective of a BFW that is coincident with a confining zone.**

The bottom of the confining shale is located at depth of 3200, which is 900 ft deeper than the BFW. The protection of the BFW is ensured by two different techniques, both numerical simulation and well tests/direct measurements as follow:

**1- Numerical simulation**

- Fracture simulation: 3D fracture simulation study was conducted to confirm the fracture is contained within the current injection sand and no fracture break out of zone. The fracture simulation results showed that the fracture will always propagate laterally within the current sand, no upward or downward propagation was predicted as demonstrated in **& Figure 30**.
- Reservoir simulation: reservoir simulation study was conducted using TOUGH2 simulator to predict the fluid migration in the formation beyond the created fracture. The reservoir

simulation study confirmed fluid migration laterally in the current sand and no flow through neither the upper nor the lower shales will occur. Please refer to **Section 3.2**.

- Well test analysis: The fall off test will be analyzed using traditional and advanced injection fall off test tools because of their capability of providing data on boundaries between multiple fluid domains.

## **2- On-going monitoring:**

- Temperature log: Temperature logs can indicate flow behind casing or tubing, such as channeling. Other applications include defining lowest point of production or injection, identification of casing, tubing, or packer leaks, and determining hydraulic fracture height (approximately at least)
- Special well test analysis: The fall off test will be analyzed using an advanced injection fall off test tool (@IPT) because of its capability of providing data on fracture containment and other critical parameters.

### **EPA comments:**

*This response, as well as the response to 6.4, does not provide information on the fracture simulation. Additionally, Figure 12 in response 3.4 indicates that the BFW is at 2,100 feet, not 2,300 feet as stated in the application.*

### **EPA questions/requests:**

- 6.7-1** Please provide more information about the fracture simulation, including the method, setup, inputs, and simulation results.
- 6.7-2** Please clarify or correct the potential discrepancy between the application, which states that the BFW is 2,300 feet, and the resistivity log in Figure 12, which indicates the base is at 2,100 feet.

## **6.8 Please describe lower confinement and clarify the mechanism for lateral confinement (e.g., is it based only on the potential for lateral fluid movement)?**

Please refer to 6.7

### **EPA comments:**

*Response 6.7 does not address lower confinement. This is significant because while the fracture simulation in Figures 29 and 30 does not show upward migration of fractures, it does show downward migration. Also, it appears that lateral migration is governed by fluid flow rather than any geologic or hydraulic features.*

### **EPA request:**

- 6.8-1** Please describe lower confinement, considering that Figures 29 and 30 do appear to show downward migration of fractures.

## 7. Attachment H, Operations Data

### Injection Rate and Pressure

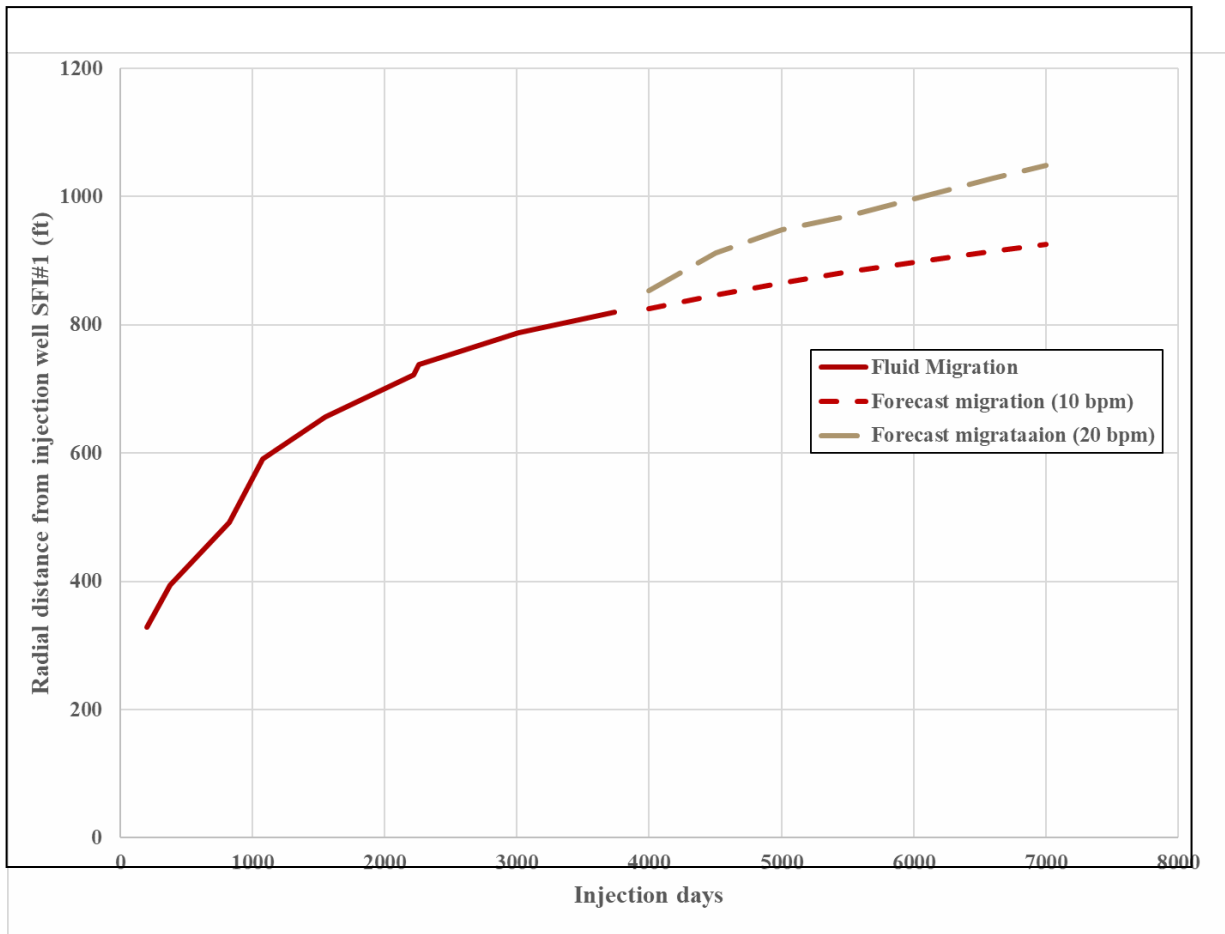
*Table 1 (page 28 of the application) shows typical injection parameters for each well.*

*The City believes that micro-seismic monitoring is becoming less useful as the project proceeds.*

#### **7.1 By how much does the City estimate that a higher injection rate and volume might expand fluid migration?**

In general, injection flow rate will only affect the near wellbore region (fracture extent) while the injection volume will control the fluid extent (flooded zone) within the injection horizon. Injection flow rate will mainly affect the lateral extent of the created fracture. the fracture simulator show that injecting at 10 bpm would create a short fracture (less than 160 ft), while increasing the flow rate to 20 bpm will extend the fracture half-length to 260 ft as shown in **Figure 29 & Figure 30**.

Based on the reservoir simulation model, the current fluid front (fluid migration radius) is 800 ft. Assuming the injection will continue at current daily volume (7,800 bbl), the fluid front will be located at 930 ft from the wellbore after 20 years of active injection at the current daily injection volume regardless the injection flow rate. However, increasing the daily volume to 19,375 bbl/day (injectingat 20 bpm for 16.18 hours a day to achieve 400 tons/day) the expected fluid front after 20 years of will be located at 1,050 ft as shown in **Figure 32**. The forecasted fluid migration with current injection rate and an increased injection rate (20 bpm) for 20 years is within the RoFD and AoR calculated in section 7.9 (shown in **Figure 35**).



**Figure 32 Forecast of radial fluid migration from well SFI#1**

**EPA comments:**

*This project entails an interplay of injection volume, injection pressure, fracture formation, introduction of biosolids, and lateral extent of fluid migration, and the information provided is challenging to synthesize as presented. As we understand it based on the City's discussion, the injection pressure increases with injection flow rate (see Figure 39 and Figure 40 below). Therefore, increasing the flow rate should increase formation pressure and fracture extent. If we understand correctly, the increased fracture volume should provide the capability to more easily accommodate the injected volume, moderating the injection pressure. Fractures could present conduits for preferential fluid flow and allow fluid to migrate farther than by matrix flow alone. Conversely, the greater volume in the fractures compared to matrix porosity could limit the lateral extent needed during injection, and leak-off into the matrix would then occur during shut-in periods. Ongoing monitoring and modeling should help better understand how the system is functioning during operation.*

*The response mentions a fracture simulator; we have not seen this simulator mentioned previously in the application, and details are not provided. (A request for information on the fracture simulations is noted above in Section 6.7).*

*The report appears to suggest that the apparent lack of additional fracturing shown in the 2017 and 2018 SRT data suggests that fluid migration is limited by a lack of new fractures.*

## **7.2 Please clarify or provide any evidence that fluid migration is limited by a lack of new fractures.**

Fluid will migrate in the formation pores of the permeable formation. In this case fluid will flow in the sand pores laterally. This is confirmed by the temperature log where no vertical fluid migration was observed. The fracture extension will not affect the lateral fluid migration; it will only affect injection pressure. A larger fracture will enable injecting more fluid at lower pressure.

### **EPA comments:**

*As noted in our comments for Section 7.1, larger fractures would allow injection at a lower pressure, but would also provide preferential flowpaths and a higher secondary permeability that would permit more rapid fluid movement laterally than through the formation matrix. While not implausible, it is not clear what information best supports the statement that fracture extension would not affect lateral fluid migration. This would ultimately need to be confirmed via ongoing monitoring and refinements to modeling.*

### **EPA questions/requests:**

**7.2-1 What information supports the statement that fracture extension would not affect lateral fluid migration? Is this based on modeling or historical information on system performance?**

## **7.4 By how much above 10 bpm does the City intend to increase injection rates? If the upper limit is uncertain, how will it be determined?**

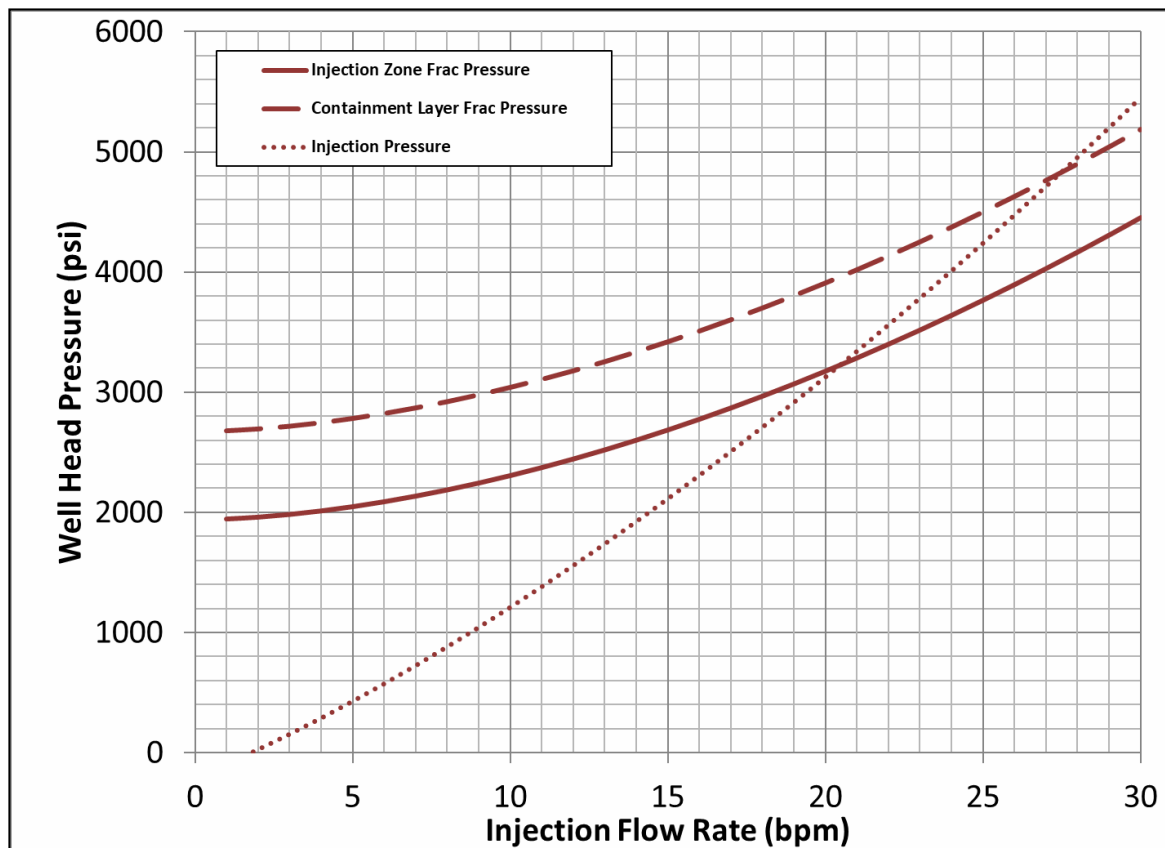
The injection pressure calculations for SFI#1 well is performed using the diffusivity equation (Equation 7).

$$BHP = \frac{162.6Q\beta\mu}{kh} \left[ \log \left( \frac{kt}{\phi\mu c_t r_w^2} \right) - 3.23 + 0.87s \right] + P_i \quad \text{Equation 7}$$

where,

$\beta$  = the fluid formation volume factor;  $\mu$  = the fluid viscosity;  $k$  = the formation permeability;  $h$  = the formation thickness;  $t$  = injection duration;  $\phi$  = formation porosity;  $c_t$  = total compressibility;  $r_w$  = wellbore radius;  $s$  = skin factor; and  $P_i$  = initial reservoir pressure.

The formation fracture pressure can be identified from a geomechanical model that has been constructed from the wireline logs. The injection pressure vs. injection flow rate calculations is presented in **Figure 33**. The calculations showed that an injection flow rate higher than 20 bpm is needed to initiate the hydraulic fracture without damaging the formation. Also, the injection flow rate should not exceed 28 bpm to avoid breaking down the containment layer. The pressure data plotted in **Figure 33** is calculated at the wellhead. The increase in the fracture pressure with increasing flow rate is due to higher friction loss in the well tubing. The bottomhole pressure plot is presented in **Figure 34**.



**Figure 33 Injection pressure calculation at the wellhead SFI#1**

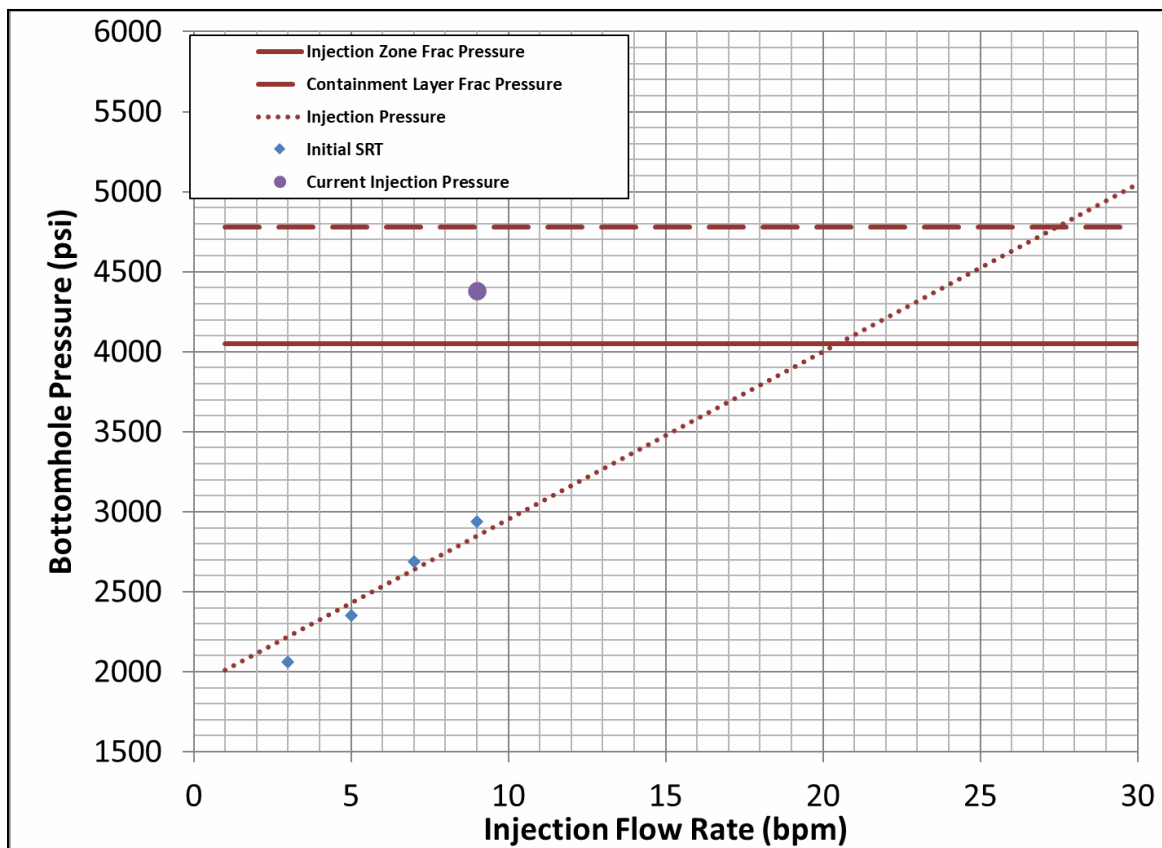
**EPA comments:**

*The application did not contain a discussion of a geomechanical model used to estimate formation fracture pressures. If the model is appropriate and well executed, it could be checked against SRT results and might provide an option to evaluate the fracture pressure of the confining unit. Without more information, however, we cannot assess the utility of the geomechanical model.*

*The method for the injection pressure vs. flow rate calculations is also not indicated.*

**EPA questions/requests:**

- 7.4-1** Please provide information about the geomechanical model used to estimate formation fracture pressures.
- 7.4-2** What is the potential formation damage that might occur at a flow rate above 20 bpm?
- 7.4-3** What relationship was used for the injection pressure vs. flow rate calculations? Is this derived from the geomechanical model?



**Figure 34 Injection pressure calculation at the bottomhole SFI#1**

### *Injection Volume*

*The application states on page 27 that “...alternating or simultaneously injecting into two wells will allow the project “...to achieve its originally proposed target injection volumes of 400 tons of biosolids per day...”*

### **7.6 How rapidly does the City intend to increase injection volume?**

Currently, four slurry batches are being injected into SFI#1 well each week. Injection is conducted from Sunday through Wednesday following the injection guidelines below:

- TIRE Injection Flow Rate: Up to the pump capacity limit (10 barrels per minute).
- Daily injection operation hours: Up to 16 hours per day (based on the bio-slurry volume available).
- Injection from a given digester ceases if the level in that digester is below 38’ feet. (subject to change pending TIWRP plant operations)
- Wet Cake Truck Loads: Average of 8 wet cake truck loads per day Sunday thru Wednesday.
- Weekly injection hours are extended (either at night or at the end of the week) if the average digester levels is higher than 52 ft (10 feet below the maximum digester levels), provided that those additional hours can be safely accommodated by TIRE (considering permit, subsurface, and surface constraints).
- Bottomhole pressure is being monitored in the monitoring wells (SFI#2 and SFI#4) to detect any communication with the injection well.

The plan is to increase the injection volume gradually depending on the biosolid volume availability to achieve the proposed injection target of 400 ton of biosolid per operating day. The daily injection

duration shall be extended gradually from 16 hours/day in 2 hours per day steps as follow (16→18→20→22→24 hours/day). Also, the injection flow rate will be increased gradually from 10 bpm in 5 bpm increments as follow (10→15→20 bpm). Finally, the weekly active days will increase gradually from 5 days/week to 7 days/week.

**EPA question/request:**

**7.6-1** *Will the changes in operation (increasing injection volume, daily duration, and flow rate) be implemented sequentially, or will more than one parameter be increased at the same time?*

**7.7 How rapidly may this affect permeability and cleanout procedures if the increased volume leads to injection of more fibrous biosolids?**

Fibrous biosolids should not affect the formation permeability since they will be trapped inside the fracture. Injecting at a higher rate results in a greater fracture volume which will reduce the solid concentration inside the fracture as shown by the fracture simulation results shown by **Figure 29 and Figure 30**. It should also minimize the need for the cleanout since larger fracture will be generated which will enhance the well injectivity.

**EPA comments:**

*The response does not appear to be substantiated. There will likely be an interplay between fracture volume and fluid and biosolids volume. If, for example, the fibrous solids accumulate along the walls of the fractures, then communication between the fractures and the matrix could be reduced by clogging. However, if the higher injection volume and greater fracture volume create greater surface area along the fracture, potential clogging might be offset.*

*It may not be possible to have a good understanding of these effects until injection has increased and data are available to assess performance.*

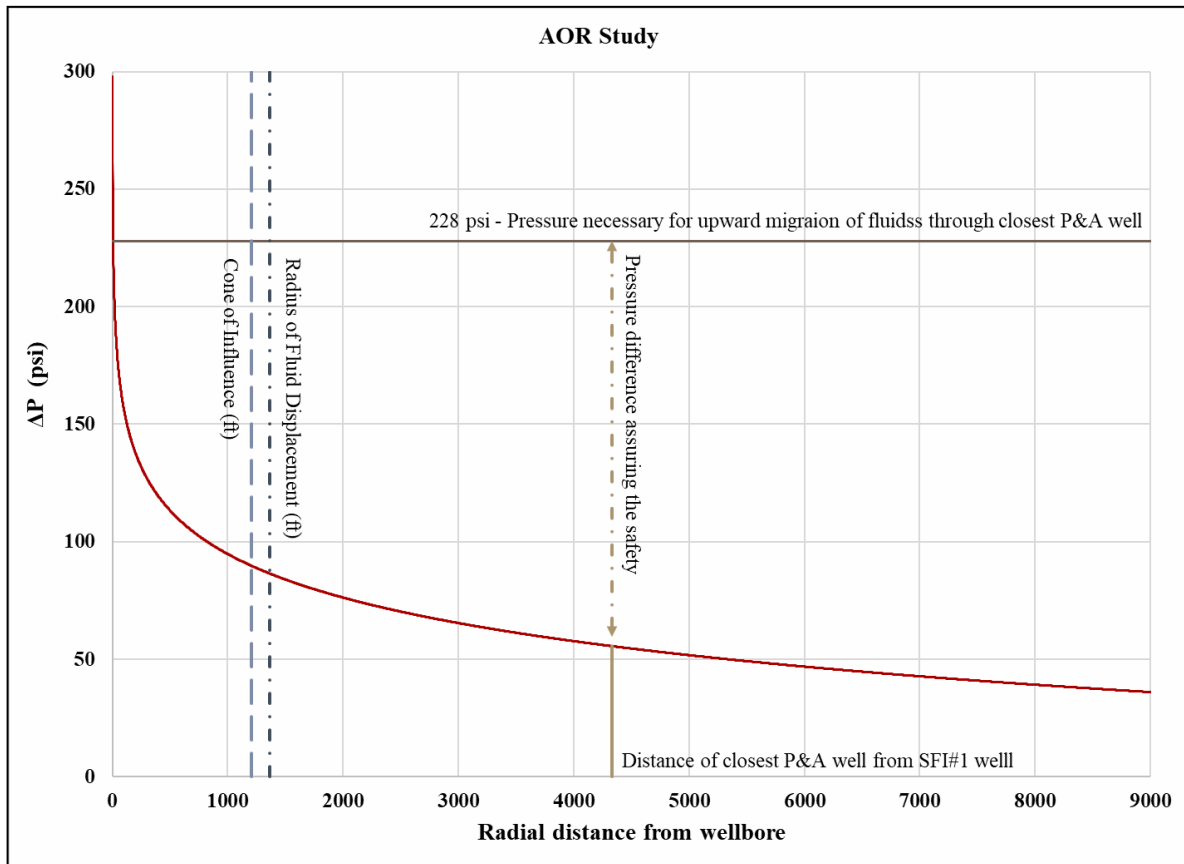
**New EPA Recommendation:**

**X.** *As changes in formation properties through fall-off testing are monitored, additional adjustments may be needed. Contingencies for necessary cleanout will be addressed in the permit.*

**7.9 How will the proposed increase in injection rate and volume and any resulting changes in BHP affect the AoR?**

The Col and RoFD is calculated using equation 1 (40 CFR 146.6) and equation 3 respectively for the current injection zone “Repetto formation” assuming injection at higher injection rate of 20 bpm. Col of 1,367 ft (**0.25 miles**) and RoFD of 1,205 ft (**0.23 mile**) indicates injection activity will not cause any disturbance to the Plugged and Abandoned (P&A) wells located more than ½ mile away from the injection well SFI#1 as shown in **Figure 35**.

After 20 years of injection operation, the pressure increase in the injection formation will be only 60 psi at 4,333 ft compared to the 228-psi required for the upward fluid propagation through the SP LA Harbor #301 well. Thus, the injection activity with higher injection rate (20 bpm) at the SFI#1 well will not compromise the USDW since the pressure increase across the radial distance is minimal.



**Figure 35 AoR around SFI#1 well assuming 20 pm injection rate**

**EPA questions/requests:**

**7.9-1 The values for Col (0.25 mi) and RoFD (0.23 mi) are different from those in Section 1.1 (page 2) (Col = 0.18 mi and RoFD = 0.1 mi for 20 years and an r of 4,333 ft.). Please clarify.**

**Shut-in period**

*The application notes on page 27 that:*

*“The City requests that shut-in period requirements be changed to allow for greater operational flexibility. Specifically, extended shut-in periods would be performed quarterly and be of a minimum duration of 20% of the cumulative injection time during the previous period. Currently the permit requires the BHP to be within 10% of the previous weeks start up BHP to move out of a shut-in period.”*

*The application also notes on page 27 that:*

*“Specifically, extended shut-in periods would be performed quarterly and be of a minimum duration of 20% of the cumulative injection time during the previous period.”*

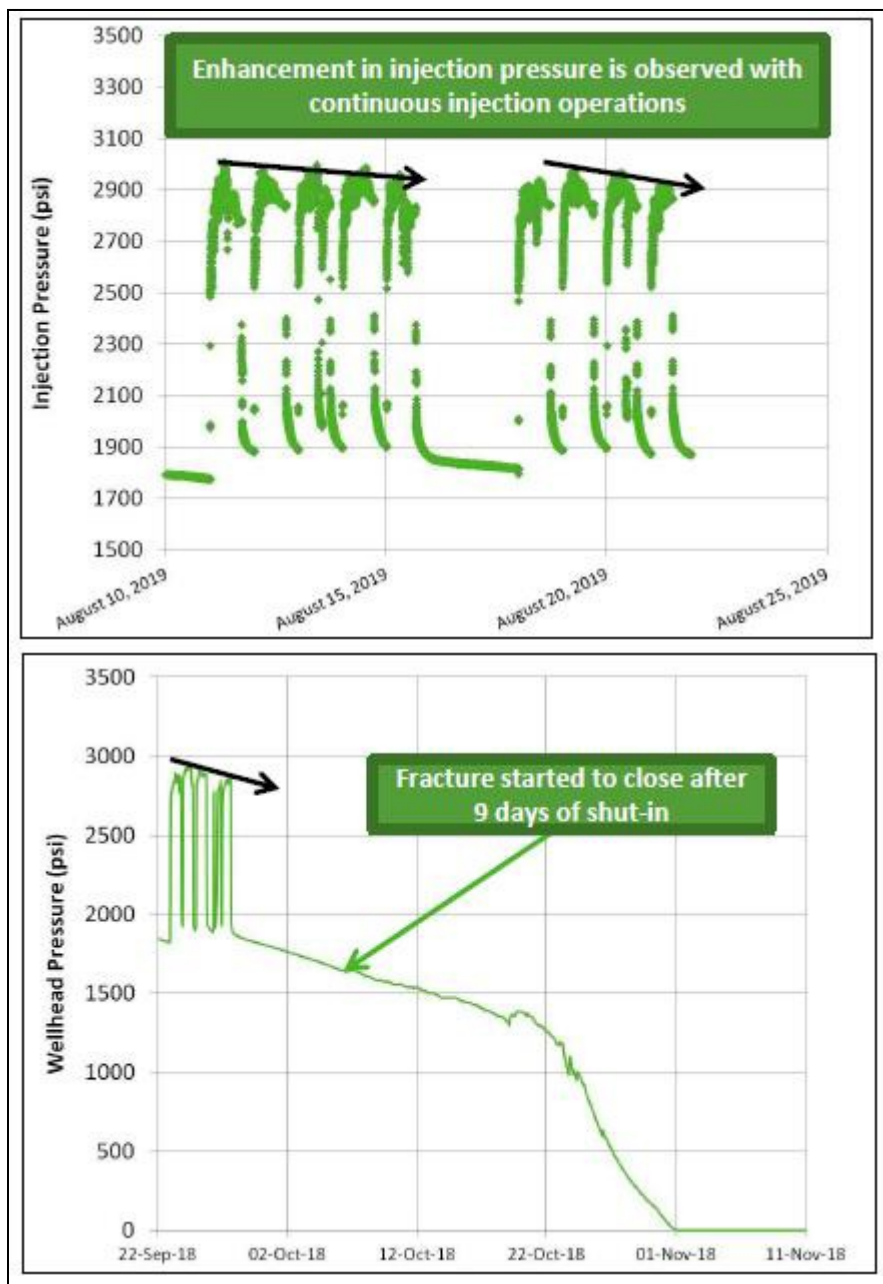
*Given the current active and shut-in periods (5 days on and 2 days off), the proposed changes could result in a significant increase in the amount of operational time.*

**7.9 How would the requested longer active periods and higher injection rates affect maximum pressures? Have simulations been done? If so, please provide and explain.**

Hydraulic fracture simulation study was conducted and presented in Section 6.4. Based on this study the predicted injection pressure at the bottomhole when injection flow rate is raised to 20 bpm will be 4,400 psi, which is equivalent to the current injection pressure at 10 bpm (up to 4,400 psi). Although the rate is doubled, enhancement in the injectivity will be observed due to larger fracture with lower solid concentration at the higher flow rate. Higher wellhead pressure is expected at 20 bpm due to the additional friction pressure loss in the well tubing when compared to 10 bpm. The calculated friction pressure loss is 376 psi at 10 bpm, and 1,235 psi at 20 bpm. Under all circumstances, the pressure will be controlled such that the bottomhole pressure does not exceed 4,780 to protect the confinement layer, as discussed in Section 7.4.

The City requests that shut in period requirements be changed to allow for greater operational flexibility. Specifically, extended shut in periods would be performed quarterly and be of a minimum duration of 20% of the cumulative injection time during the previous period. Currently the permit requires the BHP to be within 10% of the previous weeks start up BHP to move out of a shut-in period. Current shut in protocol (2-day shut-in periods for each well per week) was not very effective to allow formation pressures to decline towards baseline reservoir conditions. Also, the enhancement gained by extending the fracture during the injection operations was lost during the 48-hour shut in period and higher pressure is always observed when the injection resumes. The new proposed shut in protocol will:

1. Enhance the well injectivity by extending the injection durations
2. Allow for a longer shut in period, which will support fracture closure (extended shut in occurred on late September through early December 2018 showed that the fracture started to close after 9 days of shut in shown in **Figure 36**).



**Figure 36 Injection Pressure Decline**

#### **EPA comments:**

*The step-rate tests from 2018 show starting BHPs around 3,700 to 4,000 psi. Based on the response to this question, it appears that the BHP injection pressure would increase somewhat upon the beginning of injection but then is anticipated to stay more or less even as the project proceeds because there will be bleed-off during the prolonged shut-in times and also because the fractures are expected to moderate injection pressure.*

*The explanations scattered throughout this response document have been challenging to synthesize. If our understanding is correct, the City is essentially using a “pulsed” approach to 1) gain injectivity with the fractures during injection (also with the larger volume in the fractures mitigating the clogging effects of the biosolids), but then 2) allow them to close up during extended shut-in. The extended shut-in and associated bleed-off is expected to reduce the potential for long-term BHP buildup.*

*The response above notes that “Under all circumstances, the pressure will be controlled such that the bottomhole pressure does not exceed 4,780 to protect the confinement layer, as discussed in Section 7.4.” Figure 34 in Section 7.4 shows a confining zone fracture pressure of 4,780 psi. There does not, however, appear to be a safety factor (e.g., 80%, 95%) built into the pressure limit. The friction losses will vary with variable injection rates and injectivity over time, and the confining layer fracture pressure is uncertain without a step-rate test. A safety factor should be applied to the MASIP to mitigate the uncertainty.*

#### **EPA questions/requests:**

*The response mentions better injectivity “...with lower solid concentration at the higher flow rate.”*

**7.9-1** *Is this because it is anticipated that the injectate will be mixing with a larger volume of native pore fluid in the fractures, diluting the solids concentration?*

**7.9-2** *Please explain whether there is a safety margin built into the target maximum BHP.*

#### **7.13 How was the minimum shut-in duration of 20% of the cumulative injection time chosen?**

This basically is injecting 75 days and shut-in for 15 days each quarter. Based on the extended shut-in period occurred in 2018 the fracture started to close after 9 days of shut in.

#### **EPA question:**

**7.13-1** *Is the target BHP still no more than 10% deviation from the previous shut-in BHP? Will the injection and shut-in periods be adjusted if more time is needed to allow BHP to drop?*

#### **7.14 How will the duration of the longer active periods be chosen?**

It will be determined based on the volume of the available biosolid that needs to be injected. the fracture simulation showed that extending the injection volume from 9,000 bpd to 18,000 bpd will extend the fracture from 200 ft to 260 ft with no significant change in the solid concentration inside the fracture, which indicate no significant change in the leak off rate will be observed at larger volume.

**EPA comments:**

*The lack of change in the leak-off rate is theoretical at this point and will need to be confirmed during operations and FOTs. We also have not seen information on the fracture simulation (this is requested elsewhere in this review).*

**New EPA Recommendation:**

**XI. Permit conditions/requirements to mitigate the uncertainty in these theoretical outcomes will need to be considered.**

**7.16 How will extended shut-in times be chosen and adjusted to allow for pressure dissipation within the injection formation?**

Extended shut in times will be selected to support initiation of fracture closure and to avoid vertical fracture propagation.

**EPA question:**

**7.16-1 Will the times to support initiation of fracture closure be based on the fracture simulation model or other types of data?**

**10. Attachment P, Monitoring Program**

**Step-rate test data**

*Figure 5 and Figure 6 (Figures 13 and 14 on pages 49 and 50 of the application) show historical step rate test data. There are several ambiguities and uncertainties in these graphs. It is not stated which well these data come from. The caption for Figure 13 says that it presents data from 2008-2010, but the dates in the legend are from 2016 through 2018. Figure 14 has a legend with data points also from 2016 through 2018, but with BHP values much higher than in Figure 13 and showing a flatter slope for most of the data.*

*It is not clear that the step rate tests in Figure 13 demonstrate dual linearity or indicate a parting pressure even though the legend suggests they are from earlier years, where the earlier stage of injection might render the formation more prone to development of new fractures.*

**10.2 Please clarify which data points in Figure 13 represent data from 2008-2010. Is the legend in error?**

The legend for Figure 13 in permit is an error. The correct legend is represented in **Figure 39**. The data cover the period of 2016 through 2018 to represent the current injection sand.

**EPA comments:**

*The legend in Figure 39 below is identical to that in Figure 13 in the application, showing data from January 2016 through March 2018. The caption, however, has changed from “step rate test from September 2008 to June 2010” to “step rate test from September 2016 to June 2010.” Thus, the legend and caption still do not match.*

**EPA question:**

**10.2-1 Please clarify for each figure whether the legend is correct or the caption and provide versions with the intended (and consistent) information in both the legend and the caption. (Our current assumption is that the dates in the legend for the data points are**

*correct and that the caption would correctly read “step rate test from January 2016 through March 2018” to match the dates in the legend. Is this correct?)*

**10.3 Please clarify the meaning of the Upper Pressure Line and Lower Pressure Line in Figure 13 and how these relate to changes over time. (Are BHPs lower as time progresses and fractures expand, or is BHP expected to increase for a time as fibrous material reduced permeability in between clean- outs?)**

The Y – axis is an error it should be Wellhead pressure instead of BHP. Wellhead pressure data doesn't reflect the formation behavior (transition from matrix to fracture) due to the additional friction loss in the tubing. However, the pressure data was found to be bounded between two linear lines. The lower pressure line represents the pressure observed during the early well life as the fracture was still clean and not much fibrous material was injected. As more fibrous material injected, increase in the pressure is observed with time. The maximum pressure is represented by the upper line. The BHP is expected to increase over time due to the presence of the fibrous material within the fracture leading to low leak-off.

Early 2016 clear fracture behavior was noted as indicated by the bi-linear shape of the pressure versus rate curve (Transition from matrix to fracture is shown in **Figure 40**). After February 2016, the behavior is linear (injection tests conducted after February 2016 in **Figure 40**), indicating there is additional fracturing only in the approved layer. The EPA will be informed when a move is made to a new fracture zone. The data have been consistent over the past 18 months with no significant changes in formation stress or injectivity

***EPA comments:***

*Although it is logical that the BHP might increase over time, there are several examples in Figure 39 (revised Figure 13) where the earlier dates are near the top line (e.g., 3/4/2016, 4/4/2016), while later dates are near the bottom line (e.g., 1/20/2017, 3/4/2017, 4/27/2017). In Figure 40 below (revised Figure 14), after the first few tests showing evidence of a fracture pressure, it is not clear that there is a pattern of increased BHP or slower leak off with increasing injection rates. The data presented suggest, if anything, that the BHPs and leak off remain within a general range over time. Data generally support the concluding statement that “The data have been consistent over the past 18 months with no significant changes in formation stress or injectivity.”*

*What is uncertain, however, is whether there will eventually be enough buildup of fibrous material that injectivity will be affected. If so, this should be detectable from operations and FOTs and could signal the need to indicate to EPA that it is necessary to move to a new fracture zone.*

***EPA questions/requests:***

***10.3-1 Is it a correct interpretation to say that, after February 2016, any increases in the fracture network are due to expansion of fractures initiated in the first few tests rather than the establishment of new fractures?***

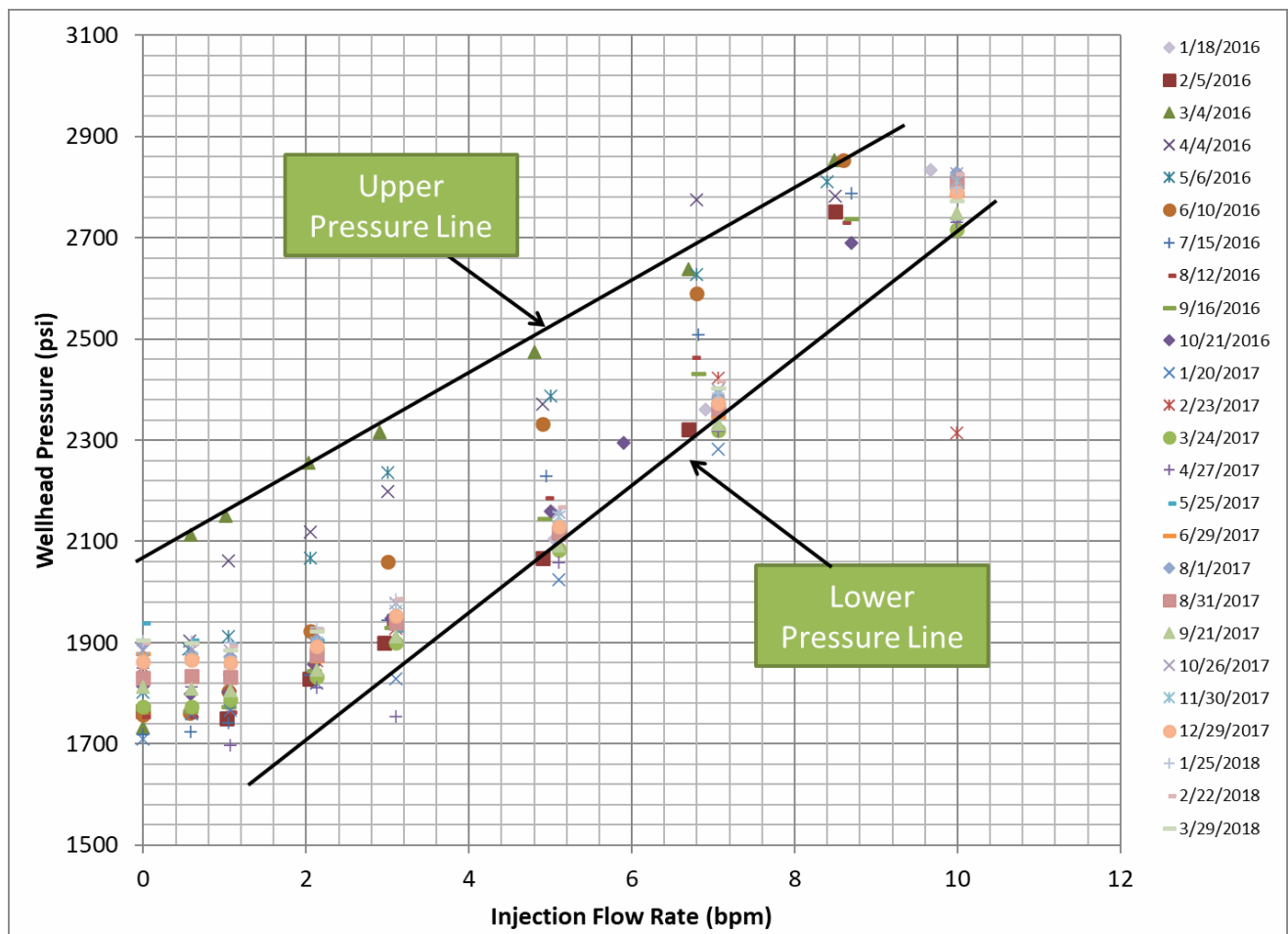


Figure 39 Step Rate Test from September 2016 to June 2010

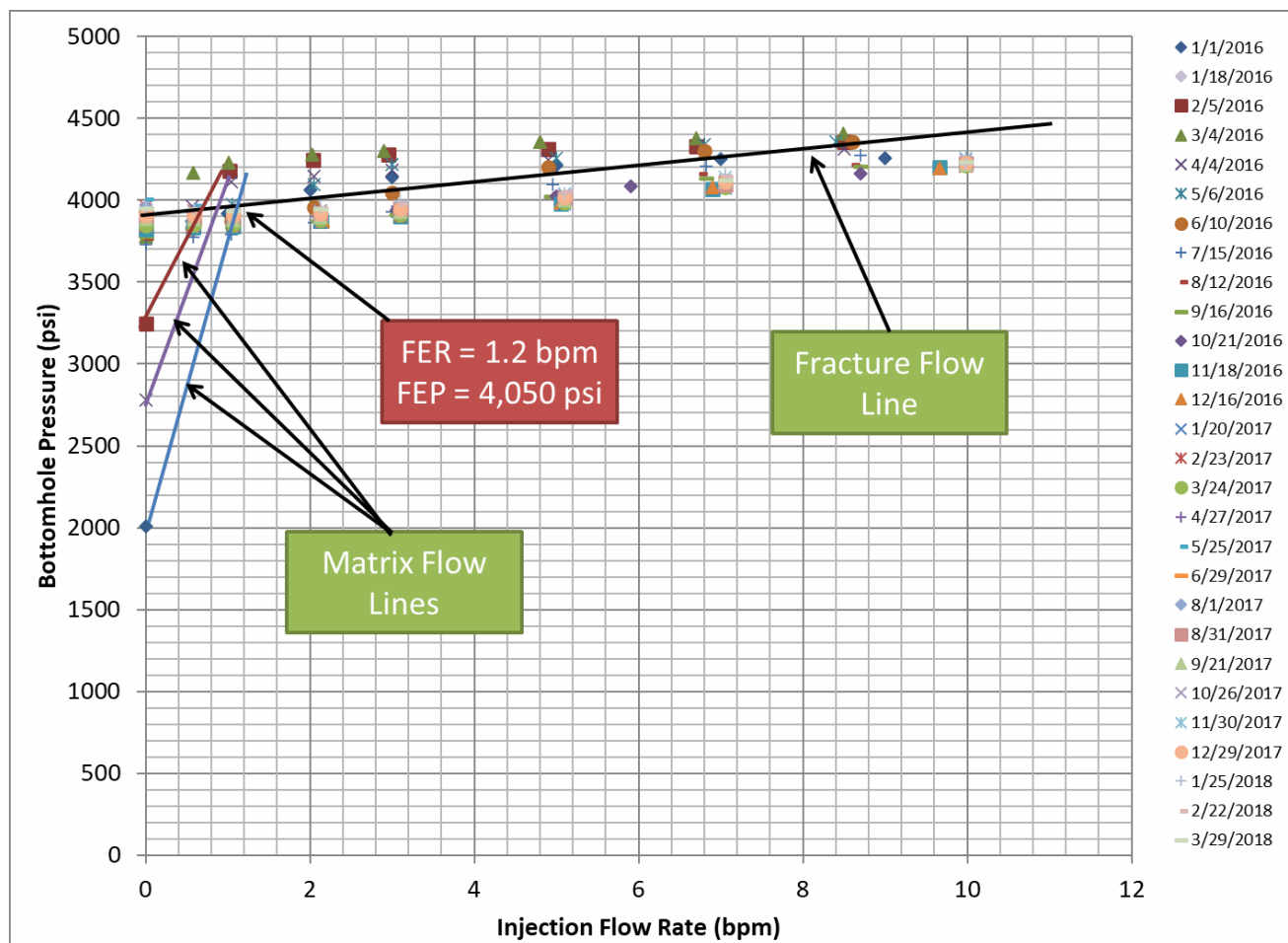


Figure 40 SRT performed with BHP sensors

**10.8** If the SRTs do not exhibit bi-linear behavior after the first few SRTs (as shown in Figure 14), what is known about how existing fractures are expanding and the implications for fluid migration?

SRT's will only show if there is a fracture or not, and the pressure at which the fracture opens. It does not provide quantitative analysis about the fracture and/or formation properties. However, the fracture growth is closely studied by performing advanced pressure fall-off test analysis. This helps in predicting the formation properties along with fracture height and fracture length and the boundaries between the inner fluid domain and the native fluid domain. Also, a reservoir simulation is performed on regular basis using TOUGH2 to monitor the fluid migration (the details to which can be found in section 3).

**EPA comments:**

**10.8-1** Please describe the advanced FOT analysis and what has been learned about fracture growth to date.

**10.9** How readily will the fractures close during off cycles with the higher pressures that would result from a higher injection rate?

Based on the extended shut-in that occurred in 2018, the fracture should be closed within 9 days of shut-in at the current injection rate (Figure 32). We expect the fracture to close in a shorter time at higher

flow rate because of larger fracture with less solid concentration will take place, which will allow the fluid to leak off to the formation faster.

**EPA comments:**

*As the fluid leaks off, the same quantity of solids will remain and could eventually clog up the sides unless the increased loading of solids is offset by the larger fracture volume and larger amount of native pore fluid to dilute the concentration of the solids. The principle being described is that the larger fractures provide an opportunity for the biosolids to be diluted. However, if injection rates are higher, then a larger total loading of biosolids may be entering the fracture network. Given there are no data to support the City's proposed mechanisms, the anticipated system performance will need to be confirmed through fall-off testing as operations proceed.*

**EPA Recommendation:**

**V. If a shaley layer in the injection formation can be identified that is lithologically like the confining zone, temporarily equipping the well to obtain data to calculate a fracture gradient may allow estimation of the fracture pressure in the confining zone without specifically compromising the confining zone itself.**

In this case, a breakdown test could be considered but could require perforation of the shale and plugging of other sand perforations which would interfere with ongoing operations. It would also lead to communication between permitted layers which could reduce ultimate capacity of the well for solids storage. Thus, this should only be considered as a test once the current perforated interval is exhausted and before perforating the new sand interval. Even then, it would be costly so the cost should be considered against the value of the data (given that there exist other less intrusive ways to estimate the shale breakdown pressure). Another option would be to conduct this test in one of the monitoring wells which would not interfere with the operation but would still be costly.

**EPA comments:**

*It is not clear what the other, less intrusive options are.*

**Questions/requests for the City:**

**10.10-1 What are the "other, less intrusive" options for estimating the shale breakdown pressure referenced in the reply? Is this referring to the geomechanical model noted elsewhere in this document?**

**10.11 Please discuss in more detail how the proposed annual thermal logging in the injection wells would replace the continuous monitoring in these wells via the fiber optic gage, to meet the objective to track vertical migration of bio-slurry.**

In addition to the annual temperature log, a quarterly advanced well test analysis will be conducted every quarter during the proposed shut-in period. Advanced well testing is achieved by analyzing fall off test data using a simulator published in the petroleum literature which factor in the formation geomechanics parameters. One such tool was published by Advantek International, the predecessor entity to the parent company of our technical partner GeoEnvironment Technologies LLC. The injection falloff test tool (@IPT) can provide data on fracture containment and other critical parameters. This advanced pressure fall-off analysis can provide better characteristic properties for the hydraulic fracture and the formation that cannot be provided by the temperature logs. Using @IPT or a similar advanced well testing tool to analyze the fall off test are as follow:

1. The use of geomechanics to allow more advanced well test interpretation including by analyzing pressure falloff data for long term fracturing injection operations
2. Provision of a more realistic estimate of reservoir and fracture parameters (length, height) through both standard well testing methods and type-curve matching of closing and shrinking fractures
3. Ability to model the shrinkage of the fracture after shut-in as it progressively recedes from containment layers and its length decreases
4. Estimation of the stress contrast between the injection and containment layers based on the shape of differential pressure and its derivative plots after shut-in
5. Ability to analyze pressure falloff data in a dual-mobility zone, resulting from difference in rock permeability and fluid viscosity.

**EPA comments:**

*According to the existing permit, dated December 23, 2013, on page 10 of 24, the existing thermal monitoring program includes two elements:*

- 1) *Real-time temperature recording in injection wells: "Injection Wells SFI-1 and SFI-3 are to be fitted with a fiber optic temperature sensor on the long string casing. Real time temperature data will be continuously recorded and displayed in the operator's control room. Reports shall graphically display monitoring results."*
- 2) *Bottom hole monitoring in monitoring wells: "A permanently installed surface recording BHP/BHT temperature and pressure gauge will be installed in the monitoring wells SFI-2 and SFI-4."*

*The objective of the thermal monitoring program, according to the June 2018 Permit Application, is (a) to understand the temperature of the reservoir to characterize the anaerobic biodegradation of the biosolids in the bio-slurry, and, (b) to track the potential vertical movement of the bio-slurry during injection.*

*Based on the explanation provided above, the proposed quarterly advanced well test analysis, including the predictive simulation/modelling and the annual temperature testing, cannot replace real time continuous temperature monitoring to detect the potential vertical migration of the bio-slurry.*

**EPA question:**

- 10.11-1** *Please describe how the proposed advanced well test analysis will detect, in real time, the potential vertical migration of the bio-slurry.*

### *Microseismicity*

*From the beginning of the T.I.R.E. project in 2006, the City of LA Sanitation Department has monitored micro-seismicity in the boreholes: in SFI #2 in 2008-2009, and 2014; in SFI #3 from 2011 to 2014; and in SFI #4 (duration not described in Appendix A). According to the August 2006 Terminal Island Renewable Energy Project Mitigated Negative Declaration (Figure 10, page 15), the monitoring would consist of tilt meters to measure fracture dimensions and continuous sensors to detect slippage. These data would support the development of fracture growth models and would ensure the 1,000- foot buffer from the Palos Verde fault. According to the December 23, 2013 UIC Class V permit, the monitoring objectives are:*

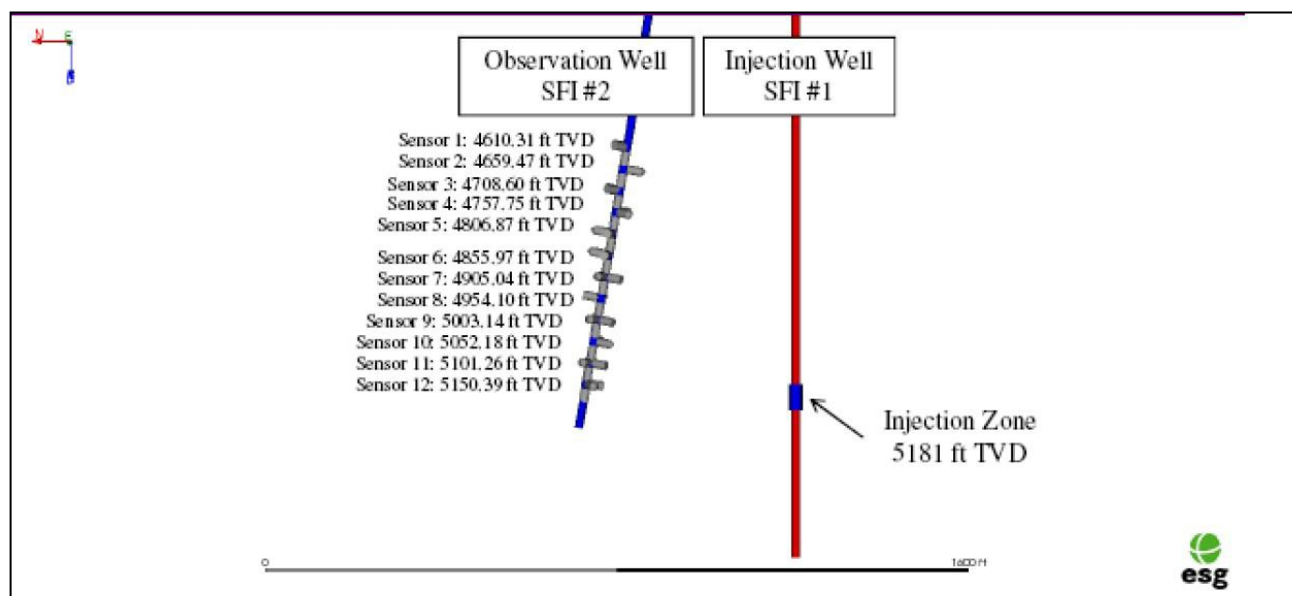
- 1) “to provide a determination of dimensions and orientation of the fractures.”*
- 2) “to provide location of seismic events related to fracturing operations within the currently authorized injection zone and within the adjacent geological confining zone.”*

*The configuration of the downhole micro-seismic monitoring array is not clearly defined in the documentation. The array is described as “12 levels of 3-component accelerometers” on pages 53 and 55 of the June 2018 UIC Permit Application with no further details. It does not describe the orientation and location of the accelerometers deployed throughout the 500-foot depth. Based on our interpretation of Figure 17 of the Permit Application (page 54), it is possible that the initial array of tiltmeters was replaced by the array of accelerometers between October 9 and November 28, 2008 (see below for the specific request for clarification).*

*Based on five letters dated November 3, 2011 through June 15, 2012 (pp. 478 through 496 of the June 2018 Permit Application) the EPA and the City of LA discussed the issue of when and how to deploy tiltmeters in the injection wells or in the proposed offset monitoring well SFI #4. This correspondence indicates that the City did not install the tiltmeters in the injection wells from the beginning of the project (although they attempted deployment in SFI#1 in 2008, 2009 and 2010) until June 15, 2012. The City notes in its letter of December 6, 2011 to EPA, that its vendor, Terralog, did not recommend deployment of downhole tiltmeters in the injection wells because there is a risk that the magnetic meters would be attracted to the injection tubing rather than to the outer steel casing (i.e., introducing a risk of decoupling). As of the letter dated June 15, 2012, it is not clear when and if the tiltmeters were deployed in the injection well or in the proposed monitoring well.*

#### **10.12 Please clarify when, if, and where tiltmeters were deployed.**

A microseismic array was installed in the monitoring well SFI#2 in mid-October 2008. This array contains 12 levels of 3-component accelerometers placed over a depth interval from 4610.3 ft to 5150.39 ft depth as shown in **Figure 41**. Nine of these sensor levels also contain temperature sensors. Data recording commenced in November, both local and regional microseismic events have been detected.



**Figure 41 Microseismic array in the monitoring well SFI#2**

**EPA comments:**

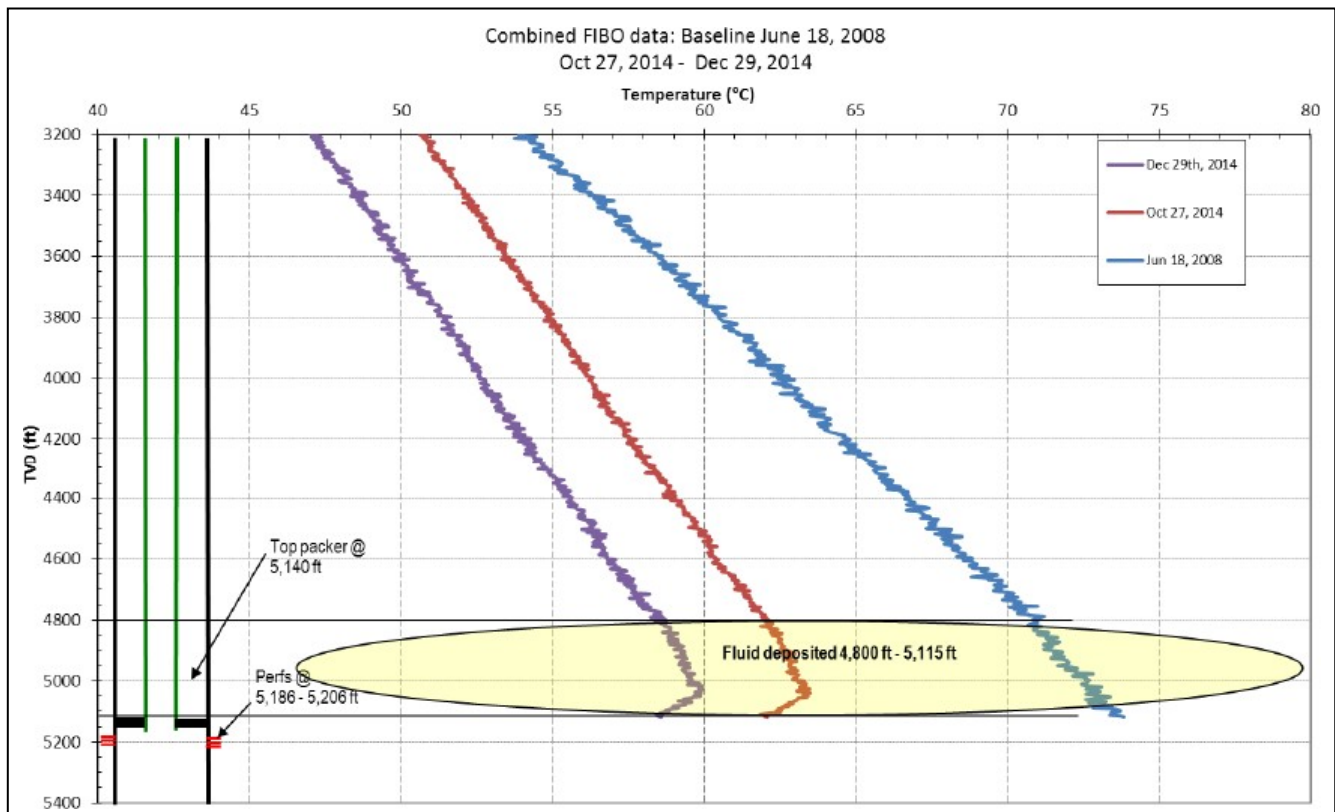
*It appears that accelerometers rather than tiltmeters were installed in October 2008. Based on the City's response to Questions 10.13 (Table 6), 10.14, and 10.18 below, the initial array of tiltmeters was not replaced by the array of accelerometers between October 9 and November 28, 2008, as originally noted in EPA comments.*

*Please see Question 10.18 below for further comments on the lack of the installation of tiltmeters.*

*Based on micro-seismic monitoring completed at the TIRE project, the City makes three claims to support a reduction in monitoring requirements in the new permit that do not appear to be substantiated by the description and data provided. Below is our assessment of each of these three claims, followed by a general summary.*

**1) Micro-seismic events measured in the overlying formation above the injection zone are not related to the TIRE operations. On page 53 of the 2018 Permit Application, the City notes that "...soft sediments like those found at the TIRE site are rather aseismic, and the vast majority of seismic events occurred in the overlying shales are due to general heaving and subsidence. Thus, the micro-seismic events are often associated with microslip and deformation in stiff lithologies, not necessarily related to hydraulic fracturing and fluid migration."**

The majority of the microseismic events occurring in the overlying shales are not related to the injection activity because the created hydraulic fracture is contained within the targeted injection formation which is confirmed by the temperature log shown in **Figure 42**. Also, the fluid that leaks off from fracture cannot migrate upwards due to the impermeable cap rock and the high stress difference between the injection zone and the overlying formation. Thus, the overlying formation is not affected by the injection activity which eliminates the chance of having a microseismic event due to injection.



**Figure 42 Temperature Profile of formation outside SFI#1 well**

**EPA comments:**

*The temperature data presented in Figure 42 do not appear to confirm that, "...the created hydraulic fracture is contained within the targeted injection formation ..." Figure 42 shows that, at a depth between 5000 and 5200 ft (i.e. the injection zone), the temperature dropped about 1-2 degrees C during the two injection events in October and November of 2014. The data collected as the baseline in 2008 do not show this temperature drop in this same range of depths. Rather the temperature in the baseline continues to increase.*

*It appears that the City is asserting that the injection zone has increased its ability to absorb or dissipate temperature (due to increased fracturing) from 2008 to 2014. If so, it is not clear that Figure 42 demonstrates that the fracture is confined within the injection formation.*

*In addition, there are no data confirming that fluids that leak as a result of the fracturing cannot migrate vertically due to the impermeable caprock, although theoretically this makes sense.*

*Finally, it is not obvious that the magnitude data presented in Figure 44 below provide sufficiently fine detail to distinguish between micro-activity occurring within the injection formation at approximately 5,000 ft., and the overlaying caprock. Figure 44 shows data grouped as deeper than 6,000 ft. and data shallower than 6,000 ft.*

*In summary, the evidence presented here does not support the claim that, "Micro-seismic events measured in the overlying formation above the injection zone are not related to the TIRE operations."*

**2) The array does not meet the monitoring objectives. On page 54 of the 2018 Permit Application, the City claims that “The microseismic array is not sufficiently precise (signals are not large enough) to track lateral fracture azimuth or lateral fluid migration.” According to the City, because the formation where the injection occurs is getting “softer,” few seismic events occur, and the “soft” formation reduces the signal so that receivers in the nearest monitoring wells (600 feet away) cannot detect and measure the signals (beyond 300 feet from the injection well, as shown in Figure 18 on page 55).**

The microseismic data was recorded for the injection activity in SFI#1 through series of microseismic array installed in monitoring well SFI#2. The data recording commenced in November 2008 and both local and regional microseismic events have been detected. The local events related to injection (i.e. events within the injection zone) range in the moment magnitude of -1 to -3 and the deeper regional events which are naturally occurring are located at depth greater than 10,000 ft and the magnitude range from -1 to +1 as shown in **Figure 43**.

Throughout the injection activity in November 2009, a decline in the local microseismic events was observed. This decrease in the microseismic events can be related to the accumulation of fibrous material like hair on the fracture face and restricting the fracture closure which is also confirmed by the Step-Rate Tests. Since no new hydraulic fracture were created the microseismic events decreased. Also as shown in **Figure 44** along with the reduction in microseismic activity the magnitude of the events has also decreased. This trend supports the interpretation that no new fracture has been created, and that the captured event with low magnitude is due to the minor event caused by a combination of the pressure difference in the formation caused due to the injection along with thermal effects (injection slurry is colder than the formation temperature).

The current injection formation “Repetto formation” is perforated within a sandstone layer which is softer compared to the overlaying and underlying shale formations. According to Warpinski, (2009), the harder and more uniform the rock, the farther sound travels and more microseisms can be picked up by the sensors. Thus, it is expected that there would be greater transmission in shale or limestone than in sandstone. Furthermore, the ongoing injection has also “softened” the formation through the introduction of more fluids into the formation. Thus, the low magnitude microseismic events cannot travel the distance necessary to away and reach the sensor. As shown in **Figure 45** the maximum distance the low magnitude events (-2 and -3) can be detected is 300 ft compared to the 600 ft distance between the injection well (SFI#1) and monitoring well (SFI#2). At a high level, microseismic analysis is a poor tool for observing fracture growth in soft, plastic sediments such as the injection zone at TIRE.

Thus, due to the above described reasons the microseismic arrays does not meet monitoring objectives.

Some technologies are being developed using ground impedance measurements which could provide an alternate route to direct visualization of the fracture geometry. That said, these impedance-based techniques are still being developed, are not typically permanently installed, struggle in the presence of large metallic objects at the surface and require relatively easy access to significant ground area.

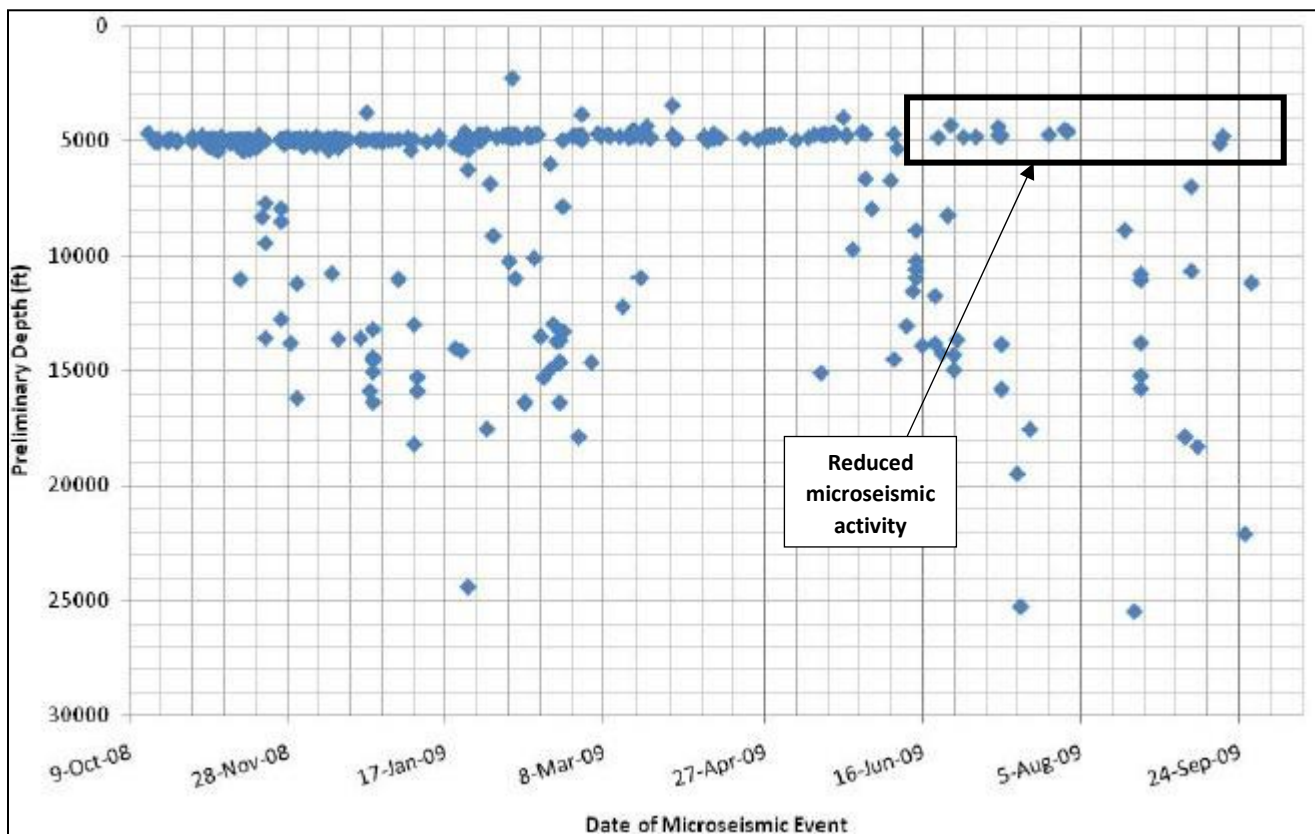


Figure 43 Depth of microseismic events with respect the injection days

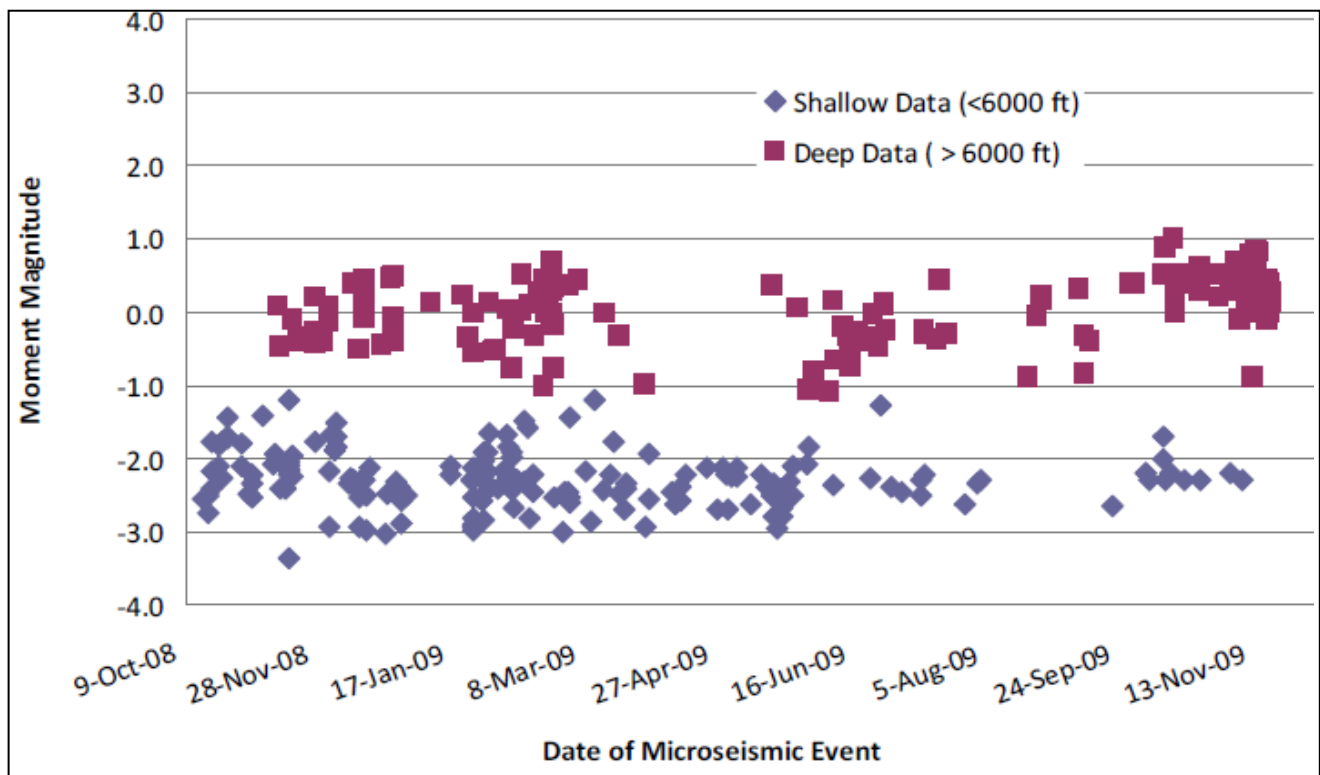
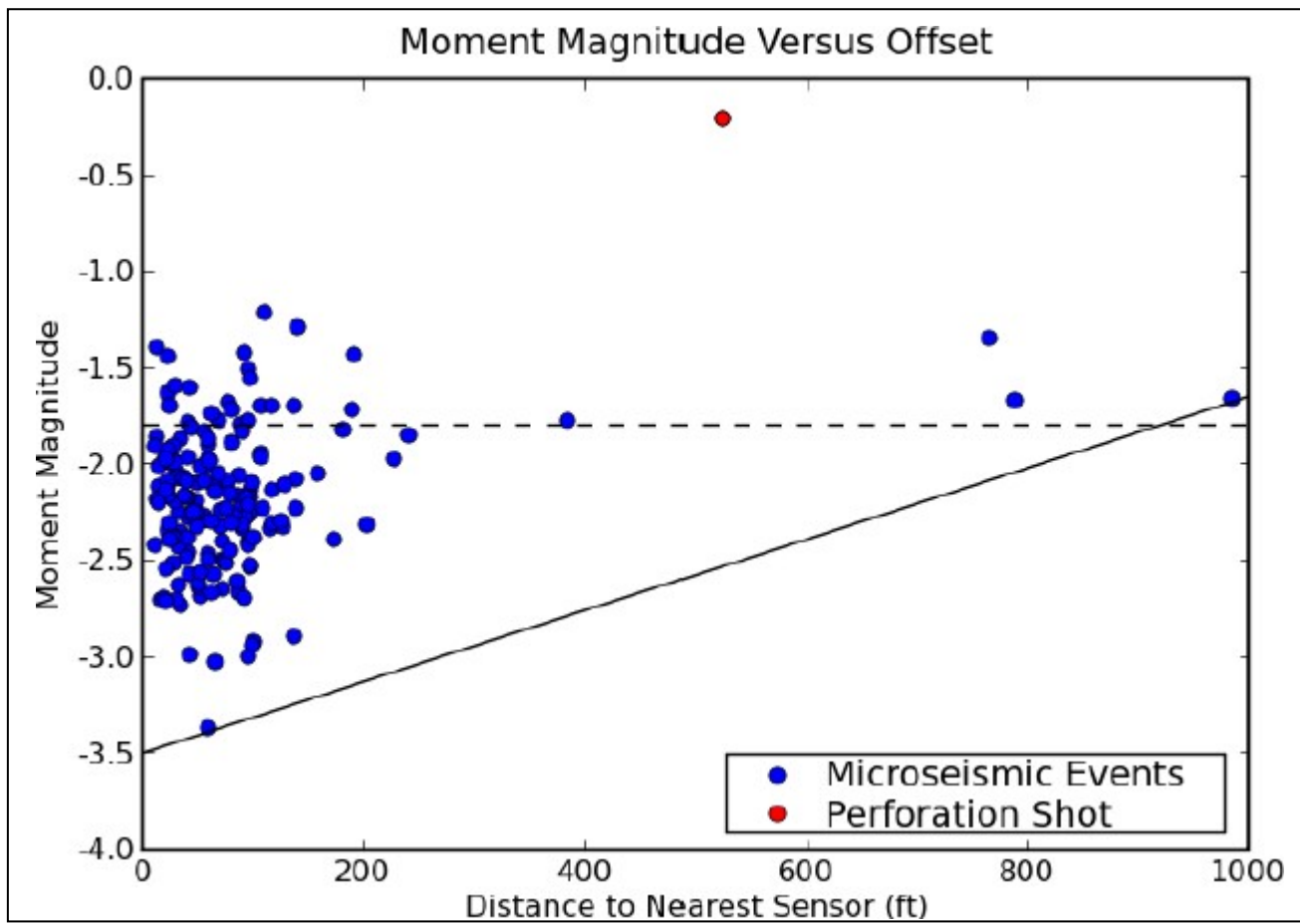


Figure 44 Microseismic events magnitudes with respect to depth.



**Figure 45 Observed moment magnitude with respect to the offset distance from microseismic array (up to September 30, 2009)**

**EPA comments:**

*Figure 43 does show fewer points around 5,000 ft. from about 28 June to 24 Sept 2009 (about 4 points over 3 months); however, this is not so clear with the deep points. It appears they are primarily below 10,000 ft., but there appear to be other points between 5,000 and 10,000 ft. Note that magnitude is not shown on Figure 43 (Figure 44 shows the magnitude of the events).*

*Figure 44 does appear to show a decrease in the number of points (i.e. micro-seismic activity), but it does not show a trend towards lower magnitude. The average magnitude in Nov 2008 seems approximately the same as in Nov 2009 (at elevations shallower than 6000 ft).*

*Regarding Figure 45, according to the City's response above, "Thus, the low magnitude microseismic events cannot travel the distance necessary to reach the sensor. As shown in Figure 45 the maximum distance the low magnitude events (-2 and -3) can be detected is 300 ft compared to the 600 ft distance between the injection well (SFI#1) and monitoring well (SFI#2)." The City's conclusion makes sense if the locations of the sensors are spaced along the distance radially from the borehole to 600 foot. However, Figure 45 may also show that most of the sensors where measurements were recorded are located within about 200 feet radially from the borehole.*

*In summary, the data presented in Figures 43, 44, and 45, do not appear to support the City's claim that, "The array does not meet the monitoring objectives." The figures appear to indicate that:*

- Micro-seismic activity at around 5,000 ft. (i.e. within the injection zone), as shown in Figure 43, or at depths shallower than 6,000 ft., in Figure 44, appears to have decreased in frequency during the 4-month interval from 28 June through 24 September 2009 (in comparison to November 2008). These data may support the City's assertion that the decrease in the frequency of microseismic events results from no hydraulic fractures during this time period. Alternately, as noted above, "This decrease in the microseismic events can be related to the accumulation of fibrous material like hair on the fracture face and restricting the fracture closure. This is also confirmed by the step-rate tests. Since no new hydraulic fracture were created, the microseismic events decreased."*
- Micro-seismic magnitude appears to have remained essentially the same from November 2008 to November 2009, at depths shallower than 6000 feet in Figure 44, which appears to contradict the City's assertion that the magnitude has decreased and does not appear to support the conclusion that, "no new fracture has been created, and that the captured event with low magnitude is due to the minor event caused by a combination of the pressure difference in the formation caused due to the injection along with thermal effects (injection slurry is colder than the formation temperature)."*
- The data in Figure 45 may indicate, either, as noted by the City, "...the maximum distance the low magnitude events (-2 and -3) can be detected is 300 ft compared to the 600 ft distance between the injection well (SFI#1) and monitoring well (SFI#2)," or, alternatively, that the majority of the sensors where measurements were recorded are located within approximately 200 feet radially from the borehole, which does NOT appear to support the City's assertion that, "...the low magnitude microseismic events cannot travel the distance necessary to reach the sensor."*

**3) Of the 40 events with micro-seismic characteristics measured at the array installed in SFI #2, during injection at SFI #1 and SFI#3 from Sept 19 through October 10, 2014, thirty-nine (39) indicate mechanical activity at injection well SF #2 (from the two calibration string shots in SFI #4 the same day), while the single remaining event is apparently due to natural seismicity.**

The summary of microseismic event is shown in **Figure 46**. These events were recorded **after** deploying a new microseismic array into well SFI#2, as the well SFI#3 was considered to be used as an injection well. There were 40 events exhibiting microseismic characters that were processed and located during the monitoring period September 19 to October 10, 2014 as described in **Table 5**

**Table 5 Microseismic events recorded**

| # of Events | Magnitude range | Attributed to                                                                                                                                                                                          |
|-------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 39          | -1.19 to -2.97  | related to the mechanical activity in or around the SFI#2 monitoring well and generally occurred on the same day as the string shot, thus it can be attributed to vibration and activity in SFI#2 well |
| 1           | -2.54           | Deep event located at 4800 ft depth was recorded after the injection operation and string shot were concluded, so its related to natural seismicity                                                    |
| 0           |                 | Associated with injection in well SFI#3                                                                                                                                                                |

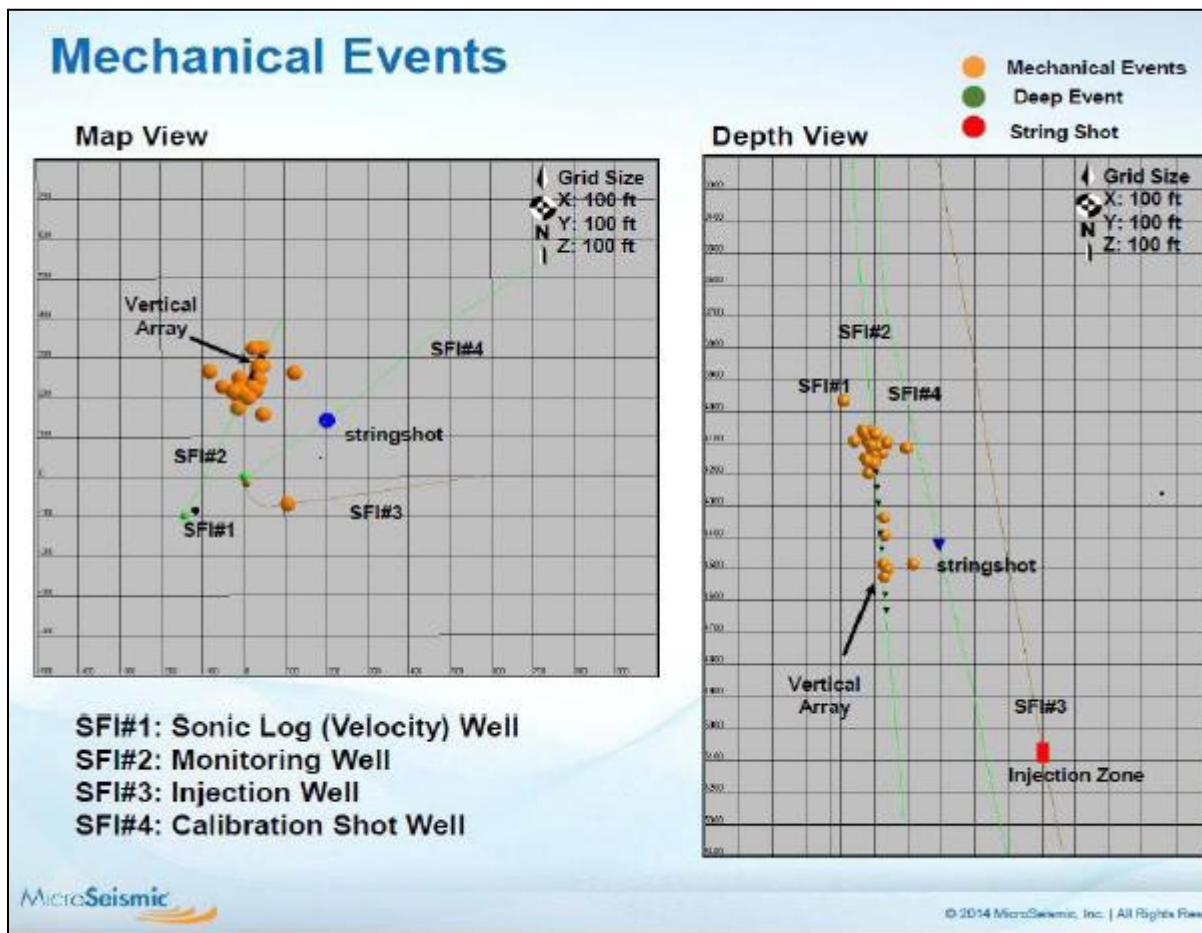


Figure 46 Map view and Depth View of the 40 microseismic events

**EPA comments:**

Over the approximately 3-week period from Sept 19 through October 10, 2014, during injection at SFI #1 and SFI#3, 40 micro-seismic events were recorded at the new array installed in SFI#2. The data for the 39 micro-seismic events (the single deepest event at 4,800 ft. does not appear in the figure) shown in the plan view and the depth view of Figure 46 do not conclusively show that 39 of the 40 events are from the stringshots in SFI#4. The map view of Figure 46 is difficult to read as the labels appear to be misplaced, and the units on the axes cannot be read. The depth view of Figure 46 appears to show that some of the events are within the same depth range as the single stringshot in SFI#4; however, the mass of the events appear to have occurred at a shallower depth, approximately 300 feet above the stringshot in SF#4. It is not clear that the micro-seismic events were NOT associated with the injection in SFI#1 and SF#3. In fact, in the depth view of Figure 46, there appears to be a recorded event at the shallower depths of SF#1 (although SFI#1 is not clearly shown). Per the City's note in Table 5, many of the events were recorded the same day as the stringshot, although a time of occurrence for each microseismic event is not shown.

Based on the information provided here, the claim that of the 40 recorded microseismic events, "thirty-nine (39) indicate mechanical activity at injection well SF #2 (from the two calibration string shots in SFI #4 the same day), while the single remaining event is apparently due to natural seismicity," does not appear to be well supported.

***Summary of EPA comments on City's three claims supporting reduction in monitoring:***

*Based on our review and evaluation of the additional detailed data provided to support the City's claims justifying a reduction in micro-seismic monitoring completed at the TIRE project, we do not find that the rationale supports these claims.*

***New EPA Recommendation:***

***XII. Based on the above, a reduction in micro-seismic monitoring requirements in the renewal permit is not supported at this time.***

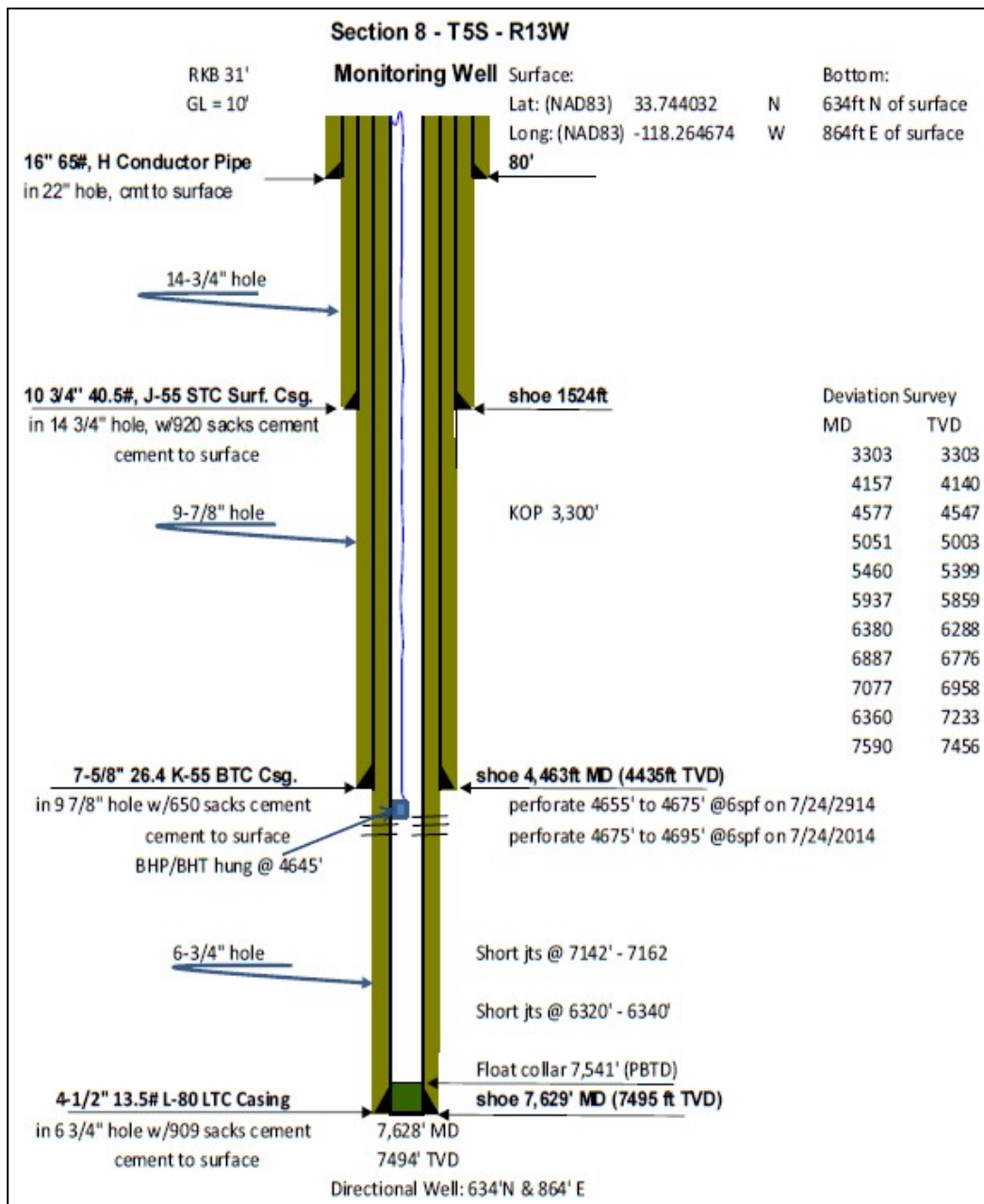
**Appendix A does not provide details on the monitoring requirements.**

**10.15 Please provide micro-seismic monitoring details.**

In addition, the construction drawing in Appendix A for SFI #4 shows the general location of micro-seismic tool, without further details.

The wells mentioned in Appendix A after the well SFI#3 are the neighboring plugged and abandoned wells.

**Figure 49** shows the wellbore schematic with detailed information.



**Figure 49 Wellbore Schematic of SFI#4**

**EPA comments:**

Figure 49 shows the details of the micro-seismic monitoring tool in SFI #4; however, the monitoring requirements have not been detailed in this response (as asserted in the original permit application).

**EPA request:**

**10.15-1** Please provide the details of the micro-seismic monitoring.

**10.16 Please provide micro-seismic tool location details.**

Please refer to **Figure 47** through **Figure 49**.

**EPA comments:**

*Please refer to the comments above.*

**10.17 Please provide additional details and a clear description of the data and conditions which support the three claims about project performance based on micro-seismic monitoring stated in the application (pages 53-55).**

Details and description of the data and conditions was provided in section 10.12.

**EPA comments:**

*Please refer to the comments above in response to Question 10.12*

*In five letters dated November 3, 2011 through June 15, 2012, pp. 478 through 496 of the June 2018 Permit Application, the EPA and the City of LA discussed the issue of when and how to deploy tiltmeters in the injection wells or in the proposed offset monitoring well SFI #4. As of the letter from EPA to the City, dated June 15, 2012, this issue was not resolved.*

**10.18 Please provide additional information to clarify resolution of this issue.**

On the microseismic, we had confirmed that the earlier data had shown levels  $\ll 1$ , and the fact that we could access existing USGS [close proximity, offset] signals/data for the injection depth for the permanent project (as a requirement for a new biosolids injection class outside the Class V).

**EPA comments:**

*This response appears to be related to another question. The response to resolution of the requirement to install tiltmeters is shown below.*

**EPA request:**

**10.18-1** *Please clarify which question this response is for.*

On the use of the tiltmeter, we stated that for the most accurate reading of the tiltmeter, it requires a "bulge" to be read. At the time, and after the opening the upper interval access "flaps", installation of a tiltmeter at the bottom of injection well seemed/perceived to read erroneous data. the "bulge" wouldn't uniformly be formed around the injection (vertically) zone, thus the tiltmeter's accuracy would be in doubt. To establish these points, GET had brought in an industry expert for a discussion with EPA.

**EPA comments:**

*As noted in the original request, and according to the five excerpts from correspondence between the EPA and the City between November 2011 and June 2012 (shown below), this issue was not resolved as of June 2012.*

**EPA request:**

*The City has provided rationales for not installing the tiltmeters, including (a) that “there is a risk that the magnetic meters would be attracted to the injection tubing rather than to the outer steel casing (i.e., introducing a risk of decoupling),” as noted in EPA’s original comments, and (b) that the tiltmeter recorded erroneous data due the lack of the “bulge” in the injection zone, as noted above.*

**10.18-2** *Please provide a discussion of resolution of the existing Permit’s requirement to install tiltmeters as noted in the letter from the EPA to the City (as described in excerpt #5 below) as it is critical to monitoring discussions for the renewal permit.*

**1) Excerpt from Nov 3 2011, EPA letter to the City of LA, page 479 of 646, from Appendix N of 2018 Permit Renewal**

existing fracturing that still remain. The City’s permit includes requirements for microseismic and tiltmeter monitoring (see Part II.C.8(f)). This monitoring is intended to provide data on the dimensions and orientation of the fracturing. To date, the City has not fulfilled these obligations of the permit. The monitoring and reporting you have provided give a strong indication of vertical containment of slurry fracture operations. However, the modeling and other analyses do not provide confidence that the dimensions and orientation of the fracturing are understood well enough. The use of tiltmeters was presented and examined in detail during the permit’s Technical Review phase and eventually approved as a permit requirement. Although the permit did not set a specific timeline for installation of tiltmeters, the equipment is considered important for obtaining the dimensions and orientation of the fracturing that has occurred to date. In particular, the permit was intended to establish, using more sophisticated means, that the minimal use of pressure and temperature monitoring are satisfactory. As a result of neglecting to use tiltmeters, questions remain unsettled regarding fracture height, azimuth and length. This information is critical to confirm that the current pressure and temperature monitoring is adequate for long term application and is sufficiently protective. This aspect of the experimental venture is particularly important given that there are no established standards and requirements, at the state or federal level. for conducting slurry fracture operations.

**2) Excerpt from subsequent letter dated December 6, 2011, from the City of LA to EPA Region 9: Downhole Tiltmeter Monitoring from the Proposed Well # 4 (SFI4)**

During late 2008 and early 2009, the City and its contractor, Terralog, made a good faith effort with an outside vendor to perform down-hole tiltmeter monitoring in the injection well (SFI1) on two separate occasions, without success. Each time there was a problem with the wireline connection and communication with the vendor’s downhole

tiltmeter array. In early 2010 we attempted a third time to install the tilmeter. The vendor informed us that they considered the injection well down-hole tilmeter monitoring "uncertain" and "not fully reliable" and they were unwilling to make a third attempt. (see Attachment A).

The City recommends that no further testing (installation) of injection well tilmeter monitoring for the following technical reasons:

1. The noted reliability problems, the inherent difficulty in deploying the downhole tilt array and measuring and analyzing data from within the injection well (there is a risk of the magnetic meters attaching to the tubing rather than to the casing);
2. Other monitoring and analysis techniques employed at the project (fiber optic temperature and microseismic monitoring) are adequately mapping vertical migration (the most important aspect of this project for groundwater protection);
3. Downhole tilmeter monitoring is not effective for fracture azimuth determination; and,
4. Finally, other measurements and analyses to date indicate that fracturing is no longer occurring from well SFI1.

To address your concerns, the City is willing to make another good faith effort to test down-hole tilmeter monitoring. To increase the chances of success, we recommend installing the tilmeter in the proposed offset well SFI4 during the start-up of approved future injection in well SFI3. The technical reasons for this approach are as follows:

1. Downhole tilmeter monitoring in an offset observation well is logistically easier than in an injection well, and the data obtained is more easily analyzed. Hence, the chances for success are better;
2. Fracturing is certain to occur during the early stages of injection into well SFI3, maximizing the chances for successful downhole tilmeter analysis; and
3. Offset well SFI4 can be optimally configured for downhole tilmeter monitoring, with no tubing and packer. Well SFI4 will be deeper than other wells, allowing the tilmeter array to completely span the anticipated fracture interval. This is very important for optimum fracture dimension analysis, and not possible with the other, shallower, injection and monitoring wells at the TIRE site.

### Tiltmeter Monitoring

In your letter and in our December 13<sup>th</sup> meeting, the City presented general information and a brief email exchange highlighting the difficulty of downhole tiltmeter monitoring, especially in an injection well. The City's letter further represents an enhanced likelihood of success with such monitoring if the tiltmeter array is deployed in a deeper, non-injection well, and at a time when new fractures are being created. The City makes the case, and EPA tentatively agrees that no new fractures are likely being created at present. However, your letter states that "measurements and analyses to date indicate that fracturing is no longer occurring from Well SFI#1." As we have noted previously, "no new fractures are being created" and "fracturing is no longer occurring" are not interchangeable statements. Note that no slurrified mixture of solid material and liquid can be injected into a porous, permeable geologic matrix such as these sand formations without either creating a fracture or propagating existing fractures. Thus, regardless of whether or not "new fractures" are being created, the injection of biosolid slurry into existing fractures is considered to be fracture injection within the UIC program (especially while operating above fracture pressure). Due to the significance of "fracturing" within the UIC program, we believe it is important that we clarify this issue, and that the City not misrepresent its injection operations in the context of the UIC program.

With specific regard to the City's proposal for tiltmeter installation, we will require some further clarification and information to properly address this issue for our UIC permit requirements. As was noted in our prior letter, tiltmeter monitoring - for the purpose of monitoring fracture growth patterns - is a very important objective of the experimental permit, and we require substantially more than the brief exchange with the vendor provided in order to make a decision on your request. For example, the communication with Pinnacle does not satisfactorily describe exactly what difficulties were previously encountered. Nor does it address whether any potential remedies might be available. Further, the City's letter states that "downhole tiltmeter monitoring is not effective for fracture azimuth determination." Yet, the City had specifically proposed the use of downhole tiltmeters and microseismic monitoring, at least in part for tracking fracture growth, when the initial permit was written. Similarly, the existing permit's monitoring/analytical plan was developed and proposed by the City and its consultants, and did not stipulate installation of a "tiltmeter array to completely span the anticipated fracture interval," nor the imperative of such an array for "optimum fracture dimension analysis" as stated in your letter. These examples are only intended to highlight some of the issues that are still outstanding and to underscore the importance of the City providing additional information regarding tiltmeter deployment for our consideration.

**4) Excerpt from City of LA letter to EPA Region 9, dated May 28, 2012, page 491 of 2018 Permit Application.**

To address your questions regarding tiltmeter monitoring, we arranged a meeting with the Chief Technology Officer for Pinnacle Technologies, Dr. Norman Warpinski, who described their technology in detail with extensive discussion on optimum deployment. During the webinar of March 12, 2012, Dr. Warpinski clearly supported the previous Terralog Technologies recommendations regarding the preferred configuration (presented to EPA in our December 2011 letter) by specifically stating that:

1. Monitoring from an offset well presented the best chance of success,
2. The tiltmeter array should span both above and below the fracture interval (see attached Downhole tiltmeter Offset Well Diagram) and,
3. Monitoring a new start-up fracture operation was more likely to have success than monitoring injection operations which have been on-going for 4 years.

Furthermore, Mr. Tony Singh of Pinnacle Technologies also added the further recommendation that tiltmeter monitoring be conducted in an “un-perforated well, absent any fluid”.

Each of these issues and the recommendations of the monitoring experts for the project (Terralog Technologies and Pinnacle Technologies) support tiltmeter monitoring in a new well (SFI4). Such a configuration will span the interval of interest, provide a dry hole condition, and allow monitoring of new fracturing with the largest possible signal. The City of L.A is supportive of the expert opinion from our project partners.

**5) Excerpt from EPA Region 9 letter to City of LA, dated June 15, 2012, page 495 of 2018 Permit Application.**

5. Your response to our letter’s discussion under “Tiltmeter Monitoring” did not adequately address our detailed concerns. We acknowledge that your preferred configuration consisting of an offset well installed with a tiltmeter array that spans above and below a new, startup fracture interval has the best chance of success. However, we believe there is an unaddressed question of whether the existing configuration will provide the azimuth of the fracture interval to a satisfactory level of understanding.

## **12. Attachment R, Necessary Resources**

*The City has provided a financial statement to EPA.*

### **EPA comments:**

*EPA continues to evaluate what the City has submitted and will provide input via separate communication.*